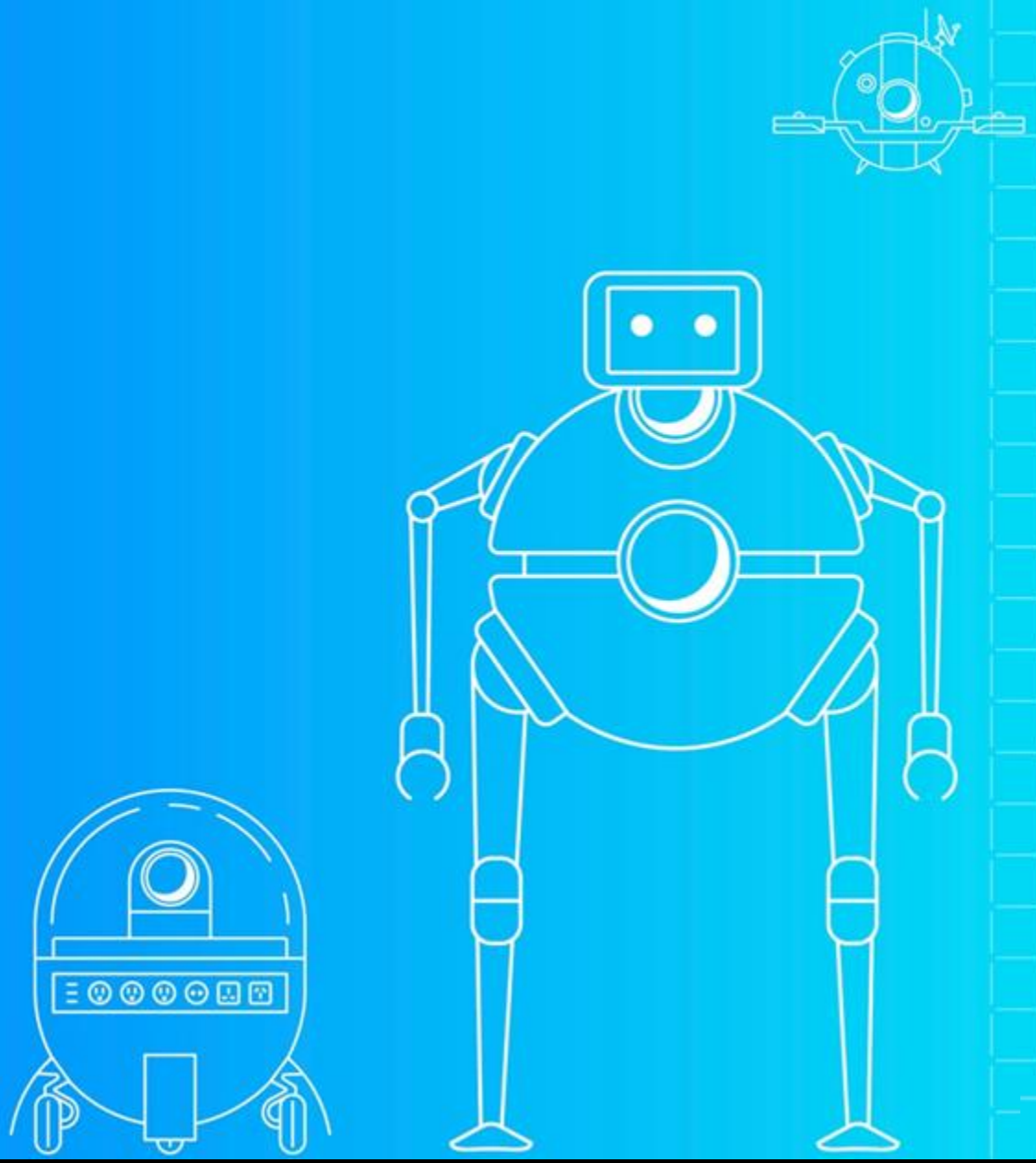
arm

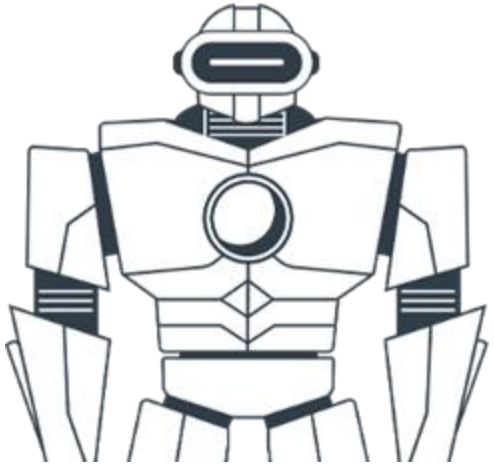
# Introduction to Arm Neoverse Compute Subsystem (CSS) and Chiplet Architecture



March 2026

# Agenda

- Introduction to Arm CPU Architecture
  - Introduction to Arm architecture
  - Introduction to AMBA specifications
- Chiplet Architecture and Arm SystemReady Program
  - Arm Chiplet System Architecture (CSA) overview
  - Introduction to Arm SystemReady for Pre-silicon test
- Arm Neoverse Compute Subsystem (CSS)
  - Introduction to Neoverse Reference Design (RD)
  - Running ACS Testing on Neoverse-RD FVP
- Enabling Chiplet Ecosystem with Arm Total Design
  - Introduction to Arm Total Design
  - Neoverse Ecosystem Partners
- Summary and Takeaway



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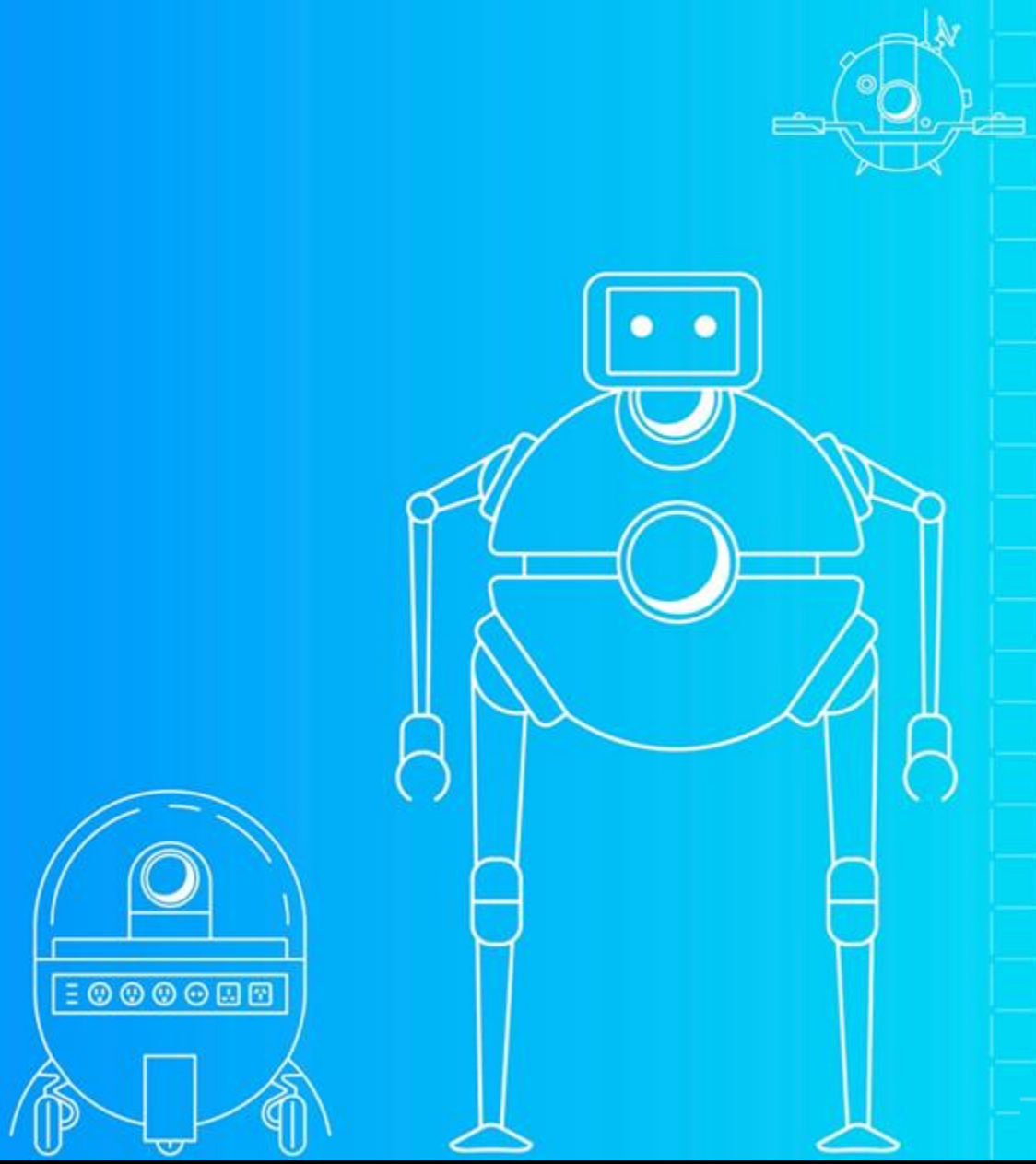
# Introduction to Arm CPU Architecture



March 2026

# What is Architecture

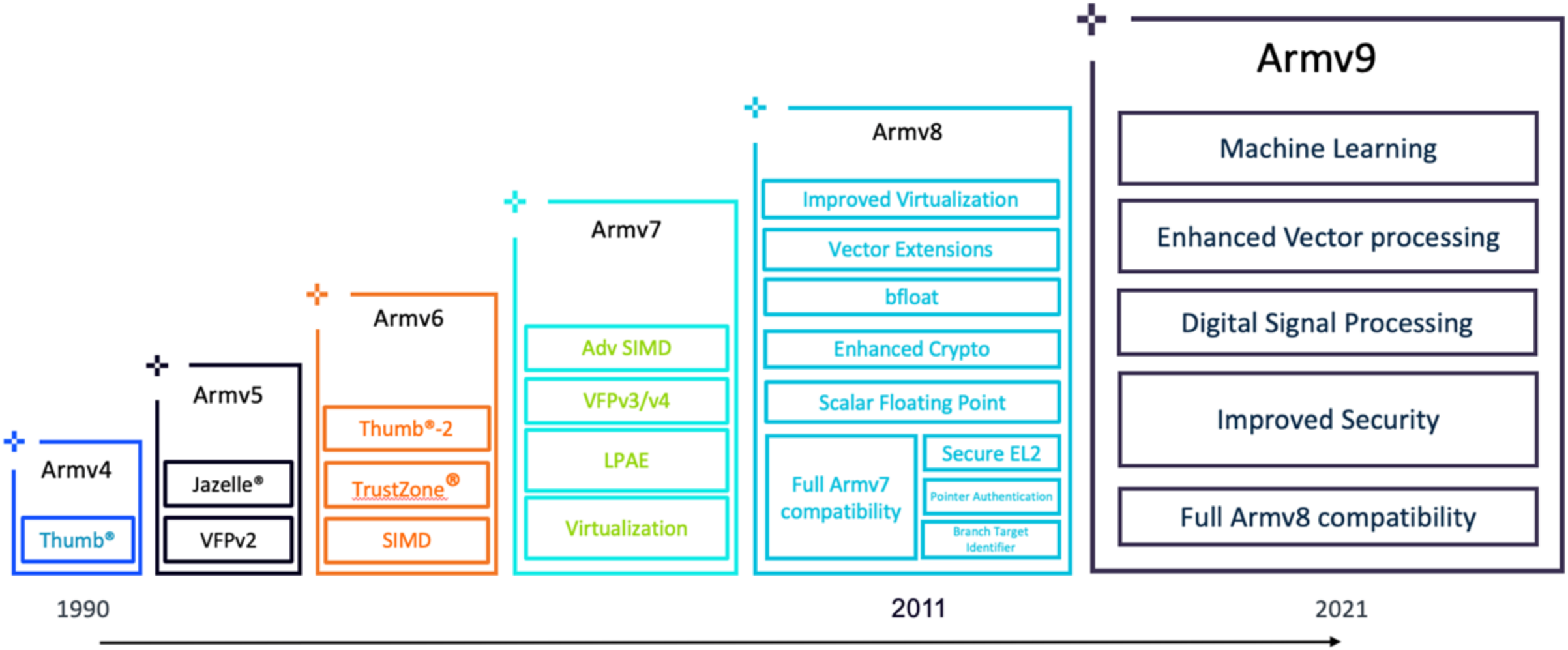
## Arm CPU Architecture - A Foundation for Computing Everywhere

- The Arm CPU Architecture defines what a CPU must do when software runs on it.
- All Arm-based CPU designs are built on the same architecture, ensuring software compatibility while enabling market or usage-specific innovation.
- Two processors that implement the same architecture can execute the same code.
- The Arm CPU architecture is implemented by a wide range of microarchitectures to deliver software compatibility across a broad range of power, performance, and area points.
  - The CPU architecture defines the basic instruction set, and the exception and memory models that are relied on by the operating system and hypervisor.
  - The CPU microarchitecture determines how an implementation meets the architectural contract by defining the design of the processor and covering such things as: power, performance, area, pipeline length, and levels of cache.

Reference: CPU Architecture

# Arm Architecture

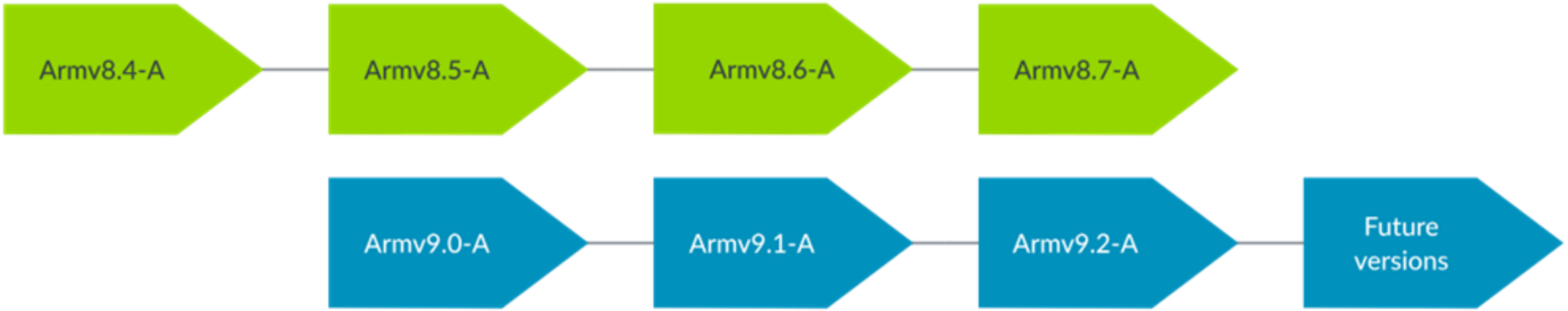
The Arm architecture has developed over time and each version builds on what came before.




Reference: [Development of the Arm architecture](), [Architecture history and extensions]()

# Arm Architecture - Annual Updates

- Armv7-A was released in 2007 and Armv8-A followed in 2011. Armv9-A was announced in 2021, nearly 10 years after first Armv8-A release.
- Because the architecture needs to evolve between major versions to add new features, minor versions, the .x extensions, are added.
- Armv9.0-A aligns with Armv8.5-A, inheriting all the features from Armv8.5-A and adding new features. After the initial release of Armv9-A, Armv8-A and Armv9-A are updated together.



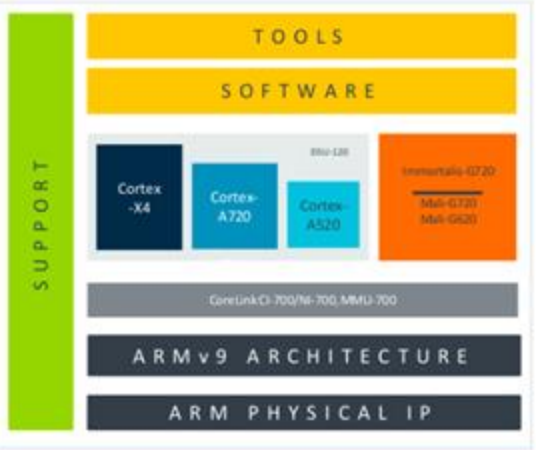

Reference: [Learn the architecture - Introducing the Arm architecture]

# Arm processors address diverse computing needs

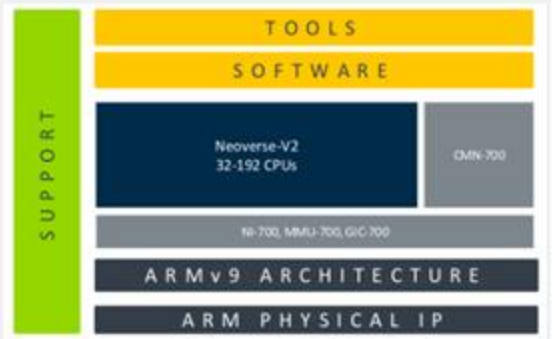
| Neoverse | Cortex-A | Cortex-R | Cortex-M | SecurCore |
|---|---|---|---|---|
| *Built for Infrastructure* | *Client, Auto, Endpoint AP* | *Faster responsiveness* | *Smallest/lowest power* | *Tamper resistant* |
| Performance, Security, Flexibility, Scalability | Designed for high-level operating systems | Designed for high performance, hard real-time applications | Designed for discrete processing and microcontrollers | Designed for physical security |

# CPU and Subsystem for Every Target End-Market

## Mobile – Total Compute Subsystem

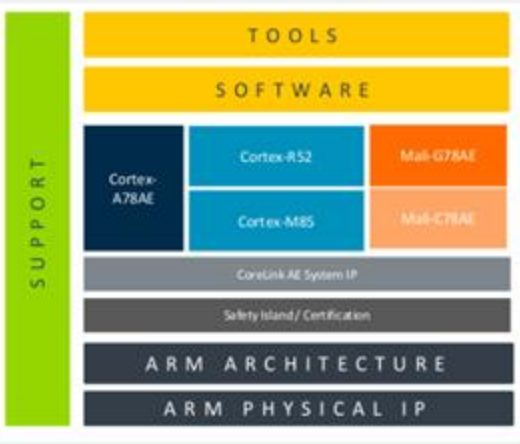


## Infrastructure - Neoverse Compute Subsystem

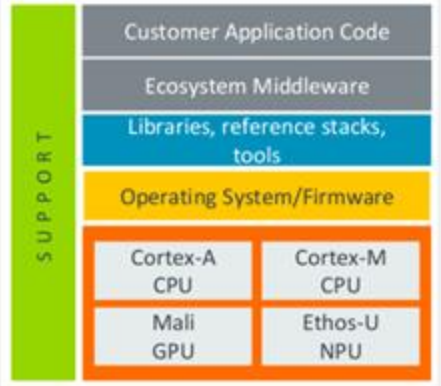



Up to
↑40%
Price/Performance

## Automotive Enhanced Subsystem



## IoT Solutions Subsystem




### Consumer IoT

Smart Speakers

Smart Cameras

Drones

Autonomous vehicles

### Industrial IoT

Industrial PC

Edge server

Industrial display

Network Hub

Smart display

# AMBA Specifications

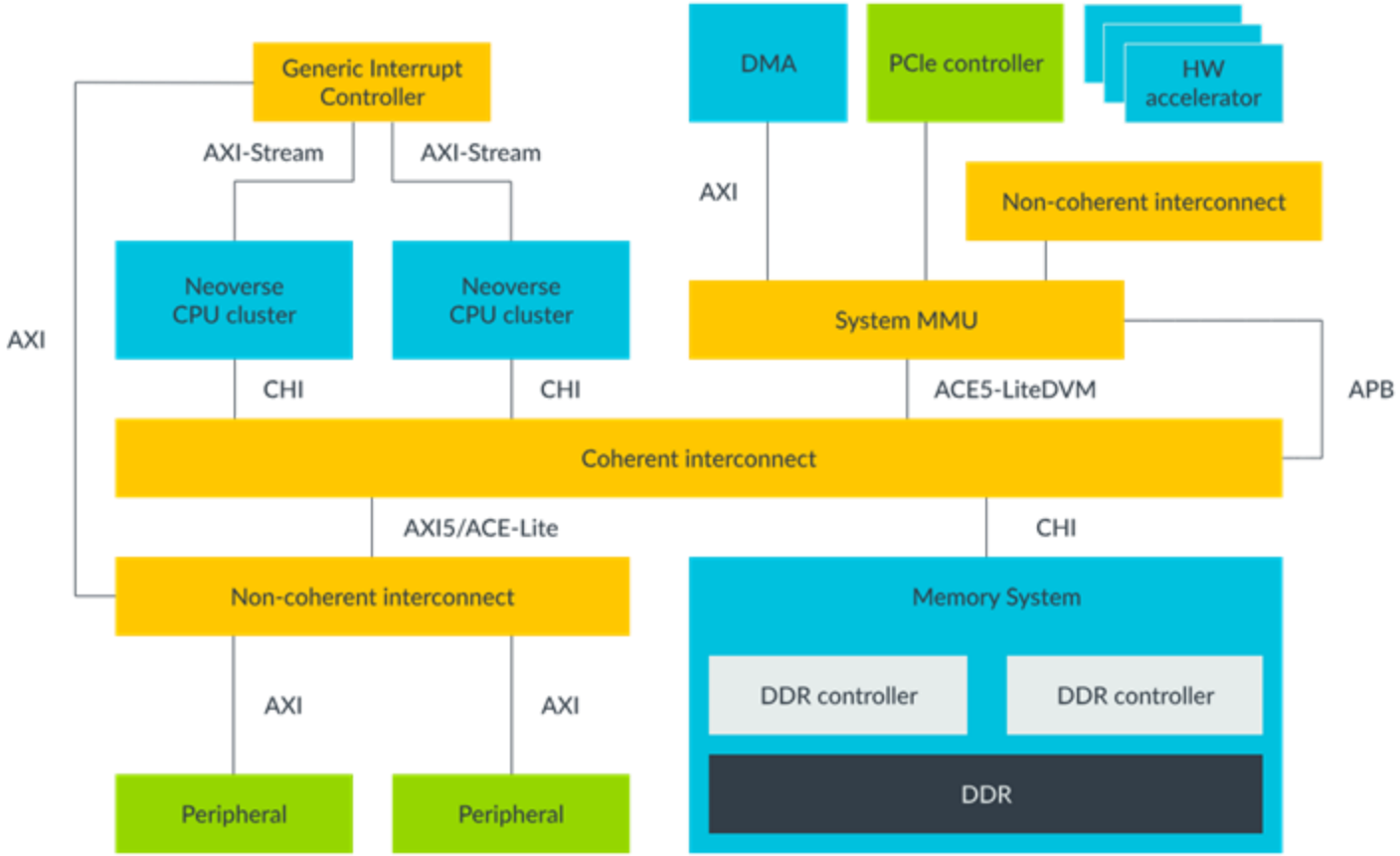
AMBA has evolved over the years to meet the demands of processors and new technologies, as shown in the following diagram:

| Key AMBA specifications | | AMBA | AMBA 2 | AMBA 3 | AMBA 4 | AMBA 5 |
|---|---|---|---|---|---|---|
| CHI<br>Coherent Hub Interface | A credited coherency protocol<br><br>Layered architecture for scalability | | | | | CHI |
| ACE<br>AXI Coherency Extensions | A superset of AXI — system-wide coherency across multicore clusters | | | | ACE<br>+Lite | ACE5<br>+Lite |
| ACE<br>Adv. eXtensible Interface | AXI supports separate A/D phases, bursts, multiple outstandings addresses, and OoO responses | | | AXI3 | AXI4<br>+Lite, +Stream | AXI5 |
| AHB<br>Adv. High-performance Bus | AHB suports 64 and 128 bit multi-manager<br><br>AHB-Lite is for single managers | | AHB | AHB<br>+Lite | | AHB5<br>+Lite |
| APB<br>Advanced Peripheral Bus | System bus for low bandwidth peripherals | APB | APB2 | APB3 | APB4 | |

Reference: [Learn the architecture - An introduction to AMBA AXI]

# What is AMBA?

The following diagram shows an example of an SoC design. This SoC has several functional blocks that use AMBA protocols, like AXI, to communicate with each other:




Reference: [Learn the architecture - An introduction to AMBA AXI]

# AMBA CHI Architecture Specification

- AMBA CHI architecture overview
  - The CHI architecture is a scalable, coherent hub interface and on-chip interconnect used by multiple components.
  - The CHI architecture allows for flexible topologies of component connections, driven by performance, power, and area system requirements.
- The components of CHI-based systems can comprise of:
  - Standalone processors
  - Processor clusters
  - Graphic processors
  - Memory controllers
  - I/O bridges
  - PCIe subsystems
  - Interconnects

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AMBA® CHI
Architecture Specification

| | |
|---|---|
| Document number | IHI0050 |
| Document quality | Release |
| Document version | G |
| Document confidentiality | Non-confidential |
| Date of issue | Mar 2024 |

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Reference: [AMBA CHI Architecture Specification]

# AMBA CHI Chip-to-Chip (C2C) Architecture

- This specification is a Chip-to-Chip (C2C) extension of the AMBA® CHI Architecture Specification
  - AMBA® CHI Architecture Specification [pdf]
  - AMBA CHI Chip-to-Chip (C2C) Architecture Specification [pdf]

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AMBA® CHI Chip-to-Chip (C2C)
Architecture Specification

| | |
|---|---|
| Document number | IHI0098 |
| Document quality | Release |
| Document version | A |
| Document confidentiality | Non-confidential |
| Date of issue | Feb 2024 |

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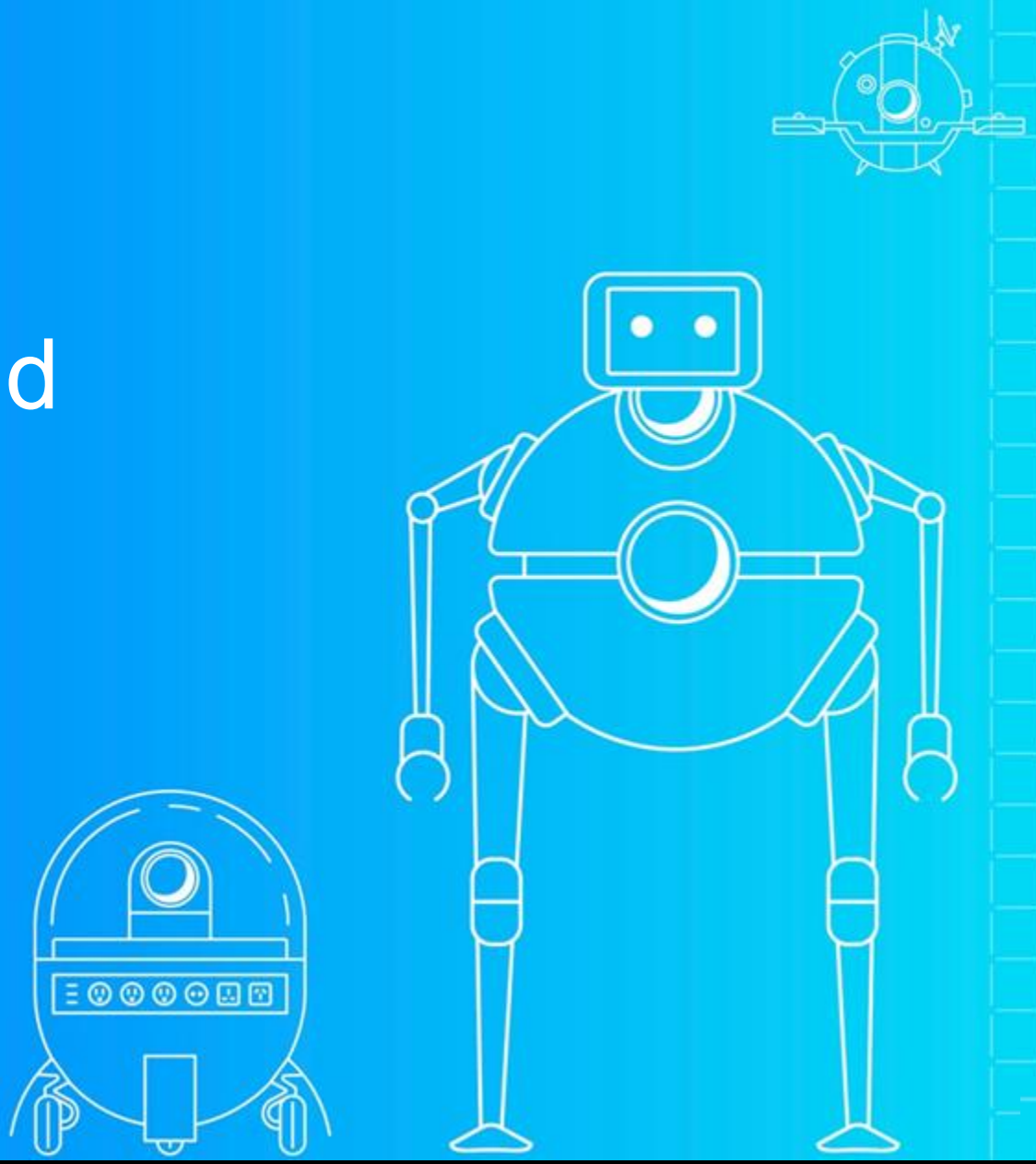
# Chiplet Architecture and Arm SystemReady



# Building Chiplets on Arm

- Arm is uniquely positioned at the center of this transformation and diverse ecosystem, with a long-standing commitment to standardization and composability.
- Through investments in tools, system architecture, and open standards like AMBA, Arm is helping to shape a flexible, multivendor chiplet marketplace that accelerates custom silicon design.

**Arm Chiplet System Architecture**

Arm and its partners are defining optimal partitioning strategies for chiplet-based systems to enable greater IP reuse across suppliers - including full chiplets.

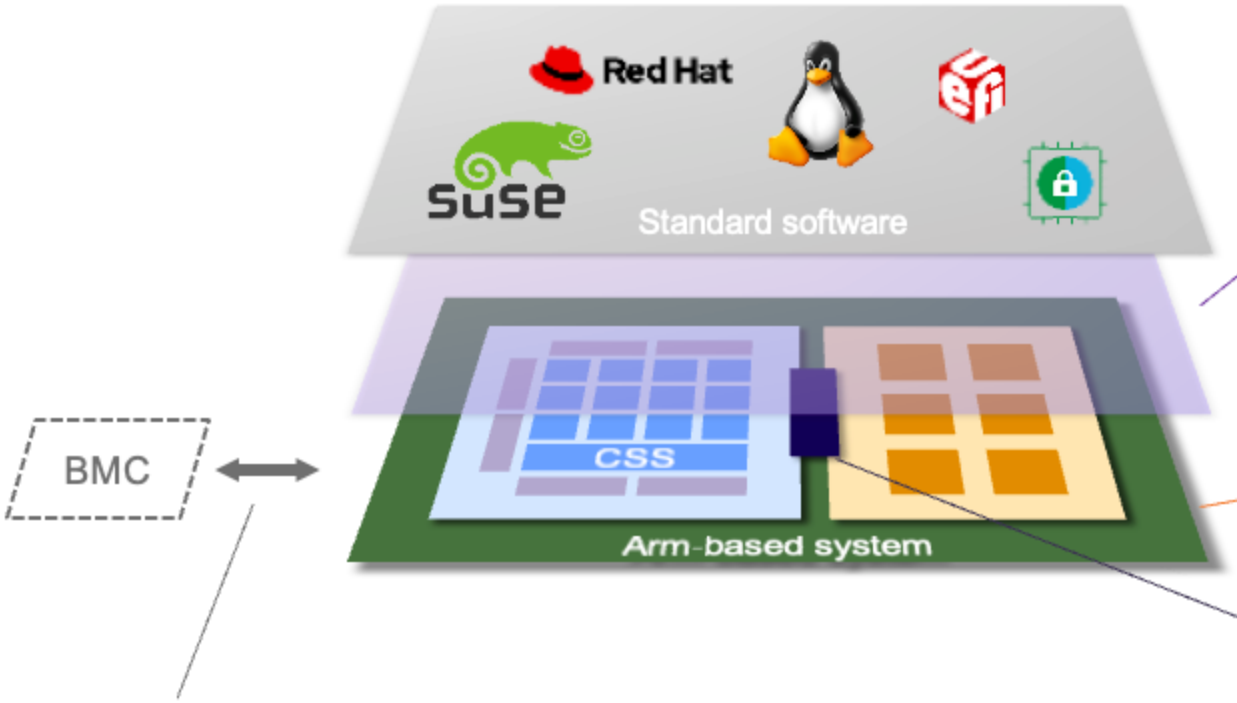
The first public CSA specification is now available, with 60+ partners applying it across markets from mobile to automotive and infrastructure.

**Powering AI with Heterogeneous Computing**

As chiplet architectures unlock new levels of performance and design flexibility, they are becoming essential for meeting the demanding compute needs of AI workloads.

The Arm compute platform is built to take full advantage of this modular approach, enabling scalable, power-efficient solutions across markets.

# Arm System Architectures Overview




## Base System Architecture (BSA/SBSA)
## Base Boot Requirements (BBR/SBBR)

- Hardware and firmware requirements for off-the-shelf software interoperability (OS etc)
- Requirements for Arm-PCIe integration

## Chiplet System Architecture (CSA)

- Partitioning of Arm-based systems across chiplets

## AMBA

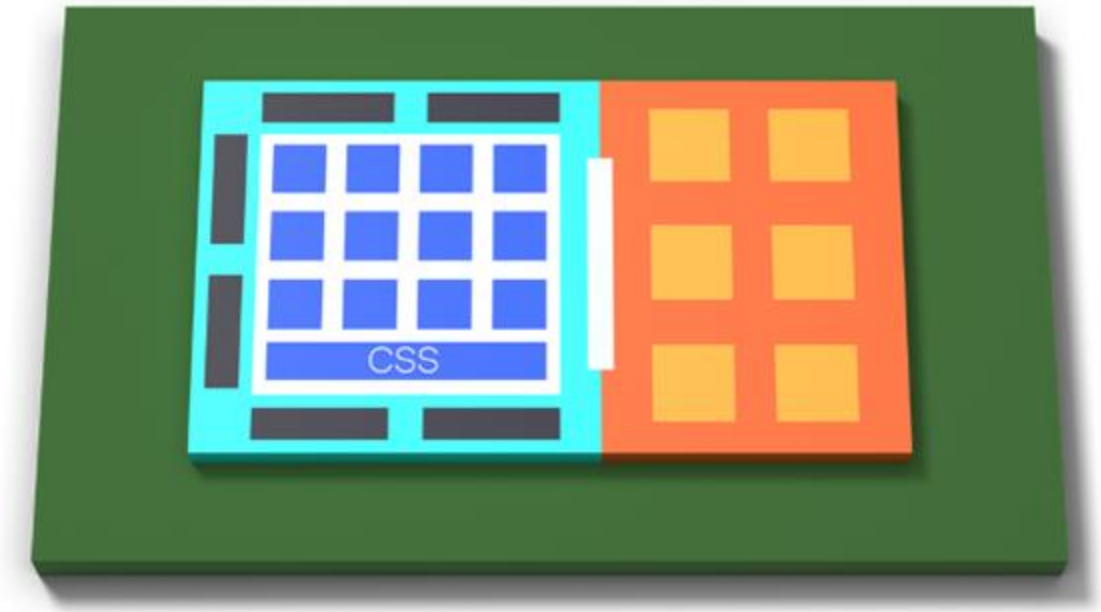
- Open standard for SoC communications underpinning silicon IP & fabless design today
- AMBA CHI C2C for connecting chip(let)-to-chip(let)

## Server Base Manageability Requirements (SBMR)

- Hardware and firmware requirements for Board Management Controller (BMC) integration

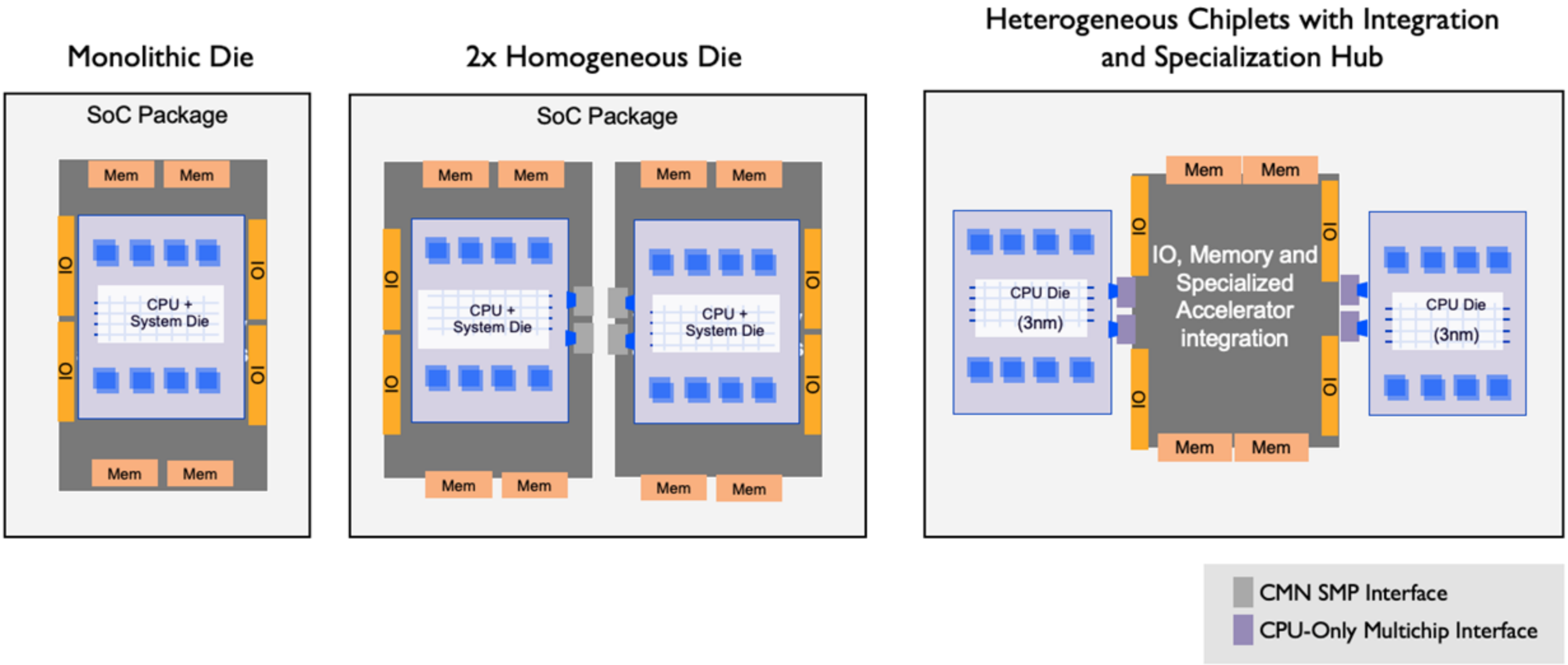
# Arm Chiplet System Architecture (CSA)

- Architectural specification for partitioning Arm-based systems into chiplets.
- Arm system designs are versatile, and the intersection with chiplets can be complex.
- The CSA provides useful definitions that simplify chiplet specification and integration.

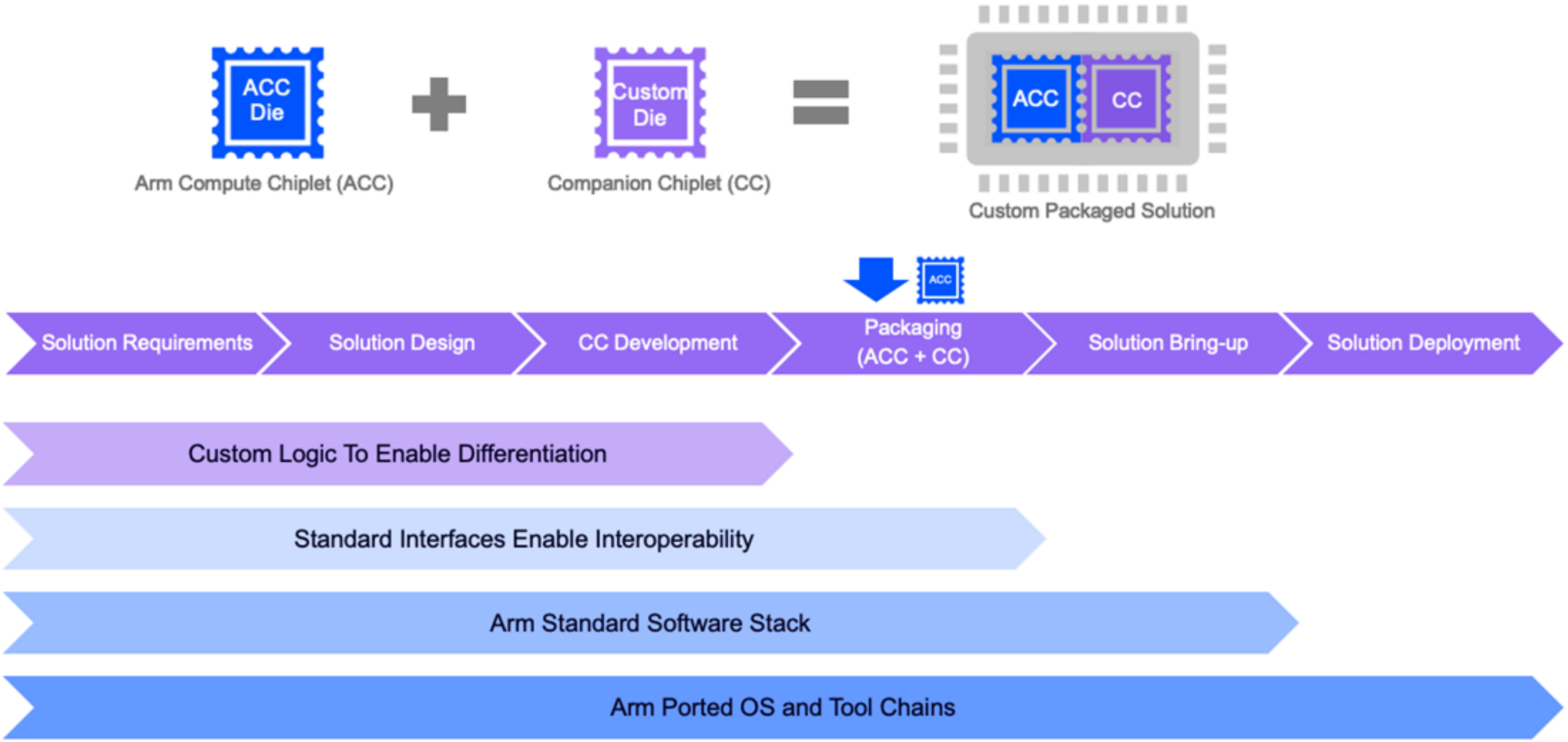



CSA – Chiplet System Architecture (DEN0145)

# Illustrated Chiplet Topologies



# Arm Compute Chiplet based Solution



# Introduction to Arm SystemReady

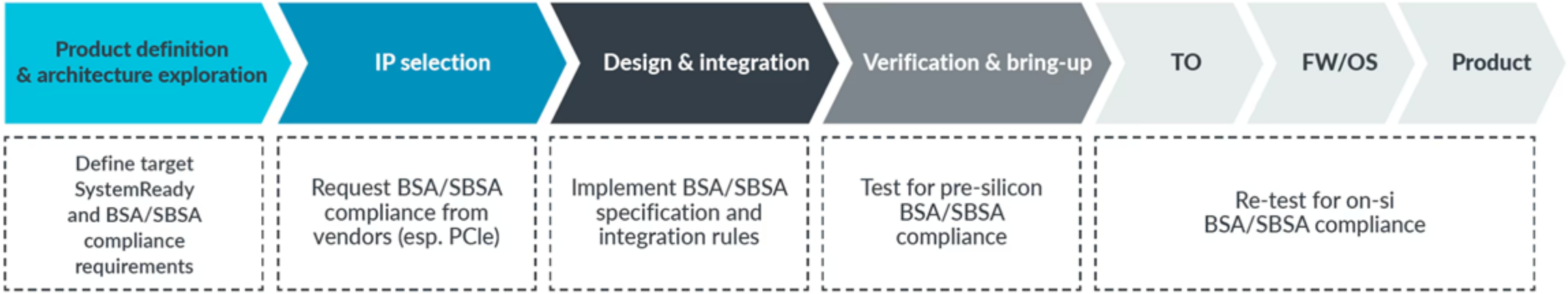
Systems that are designed to just work for the end user with the ability to install and run generic, off-the-shelf operating systems out of the box, must follow a set of minimum hardware and firmware requirements.

- **Arm Base Boot Requirements (BBR)**
  - Defines a standard set of boot and runtime interfaces provided by firmware running on Arm 64-bit Architecture application processors. Standard operating systems and hypervisors can rely on these interfaces being available.
- **Arm SystemReady Compliance Program**
  - Arm SystemReady is a compliance program that helps ensure the interoperability of an operating system on Arm-based hardware.
  - Developers can build software once and deploy it on any compliant Arm-based chip.
  - Arm SystemReady benefits the entire ecosystem as existing software can move through hardware generations and hardware providers can reach a broader addressable market.

https://github.com/ARM-software/arm-systemready

# Arm SystemReady - Pre-Silicon Program

- Arm SystemReady Pre-Silicon is a program to help silicon vendors achieve BSA compliance prior to taping out, in a well-defined and low-risk path to SystemReady.
- It provides tools (such as the pre-silicon BSA/SBSA compliance tests), and, above all, a framework with specific steps for silicon vendors to take to become BSA compliant.
  - **Guides**: SystemReady Pre-Silicon BSA/SBSA Integration and Testing Guide
  - **Standards**: Base System Architecture (BSA) and Server Base System Architecture (SBSA)
  - **Compliance Testing and Tools**: The System Architecture Compliance Suites (ACS) provide suites of tests to check that SystemReady architectural rules are understood and implemented correctly




https://www.arm.com/architecture/system-architectures/systemready-compliance-program

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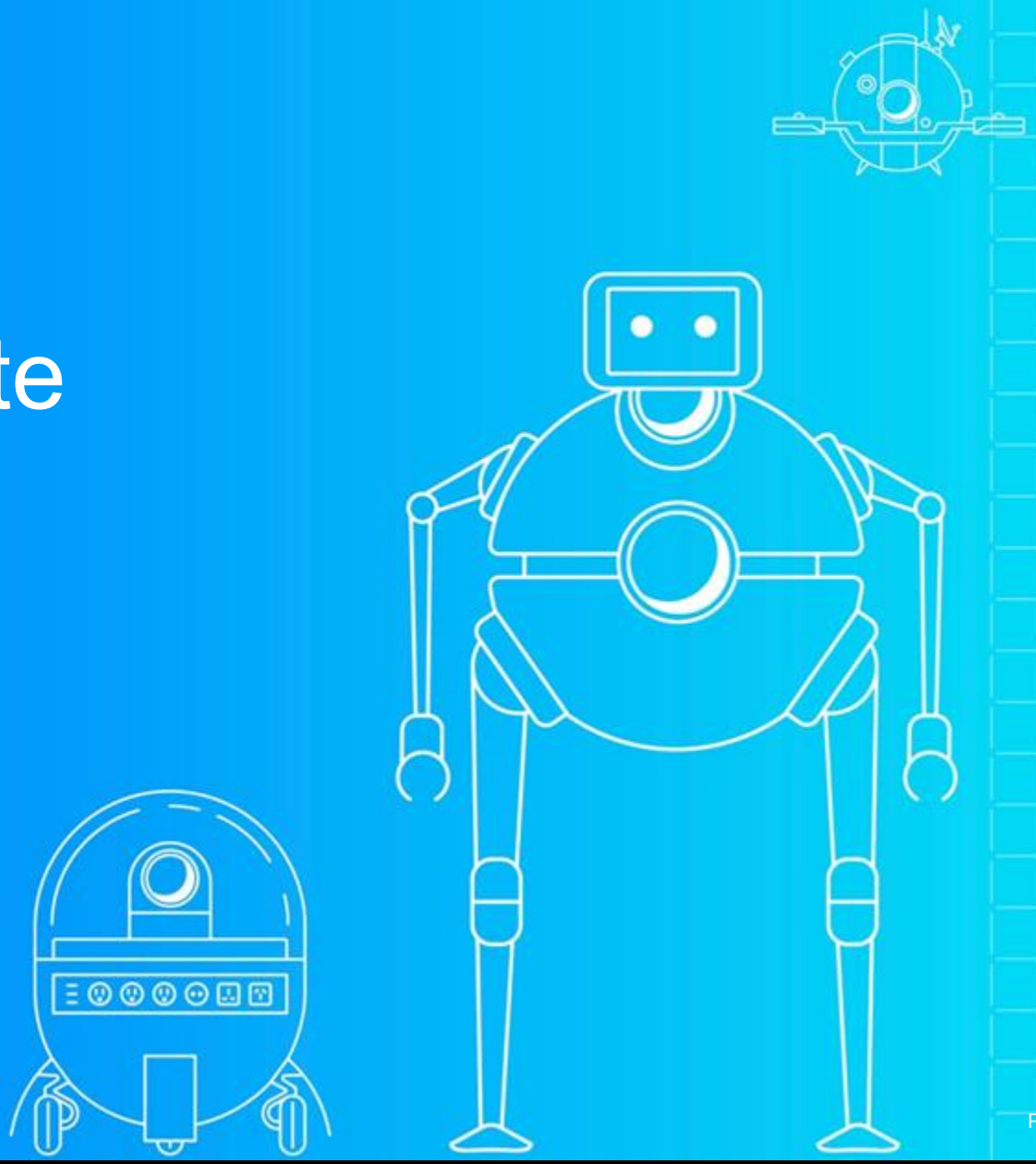
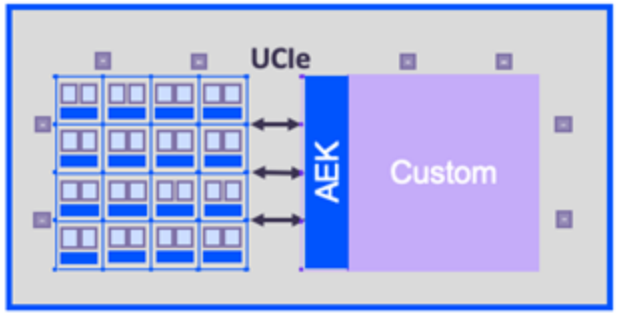
# Arm Neoverse Compute Subsystem (CSS)

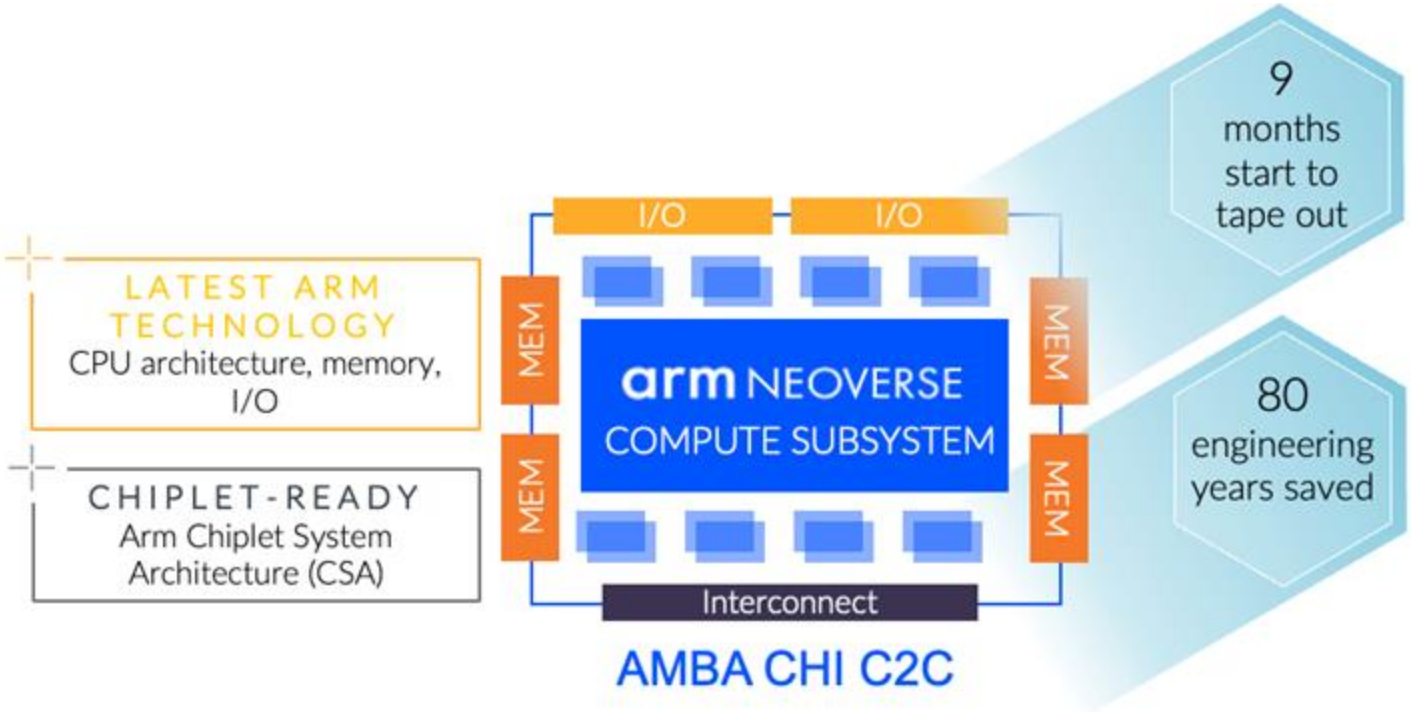


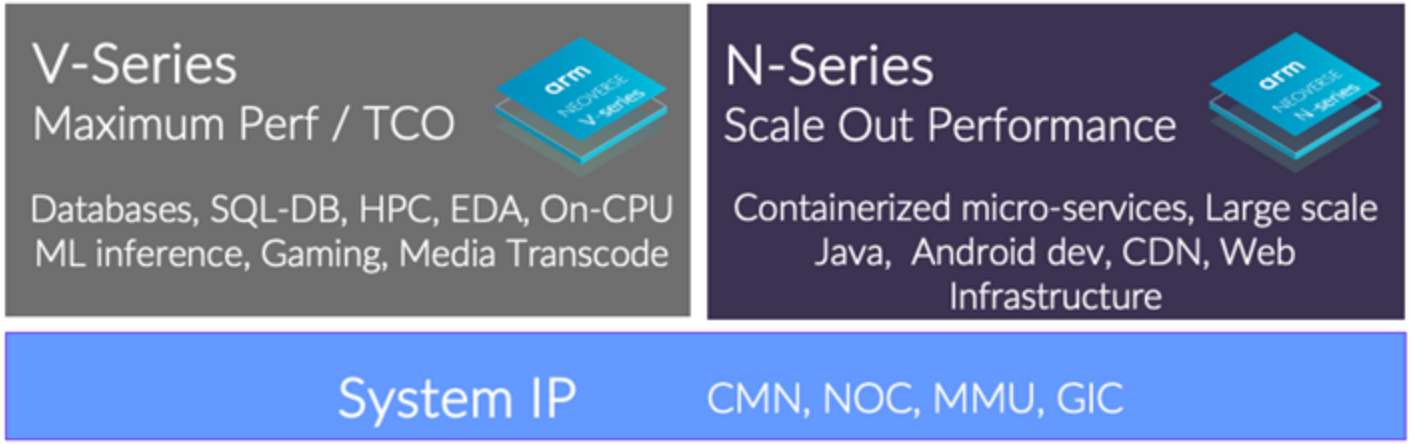
# Arm Neoverse Overview

## Neoverse Compute SubSystem (CSS)

- Pre-configured Neoverse IP with licensee input for target use cases
- Includes CPU integrated with System IP and overall SoC management
- Verified, validated, and PPA-optimized by Arm
- Delivered with reference design, implementation and Day-1 ready SW stack



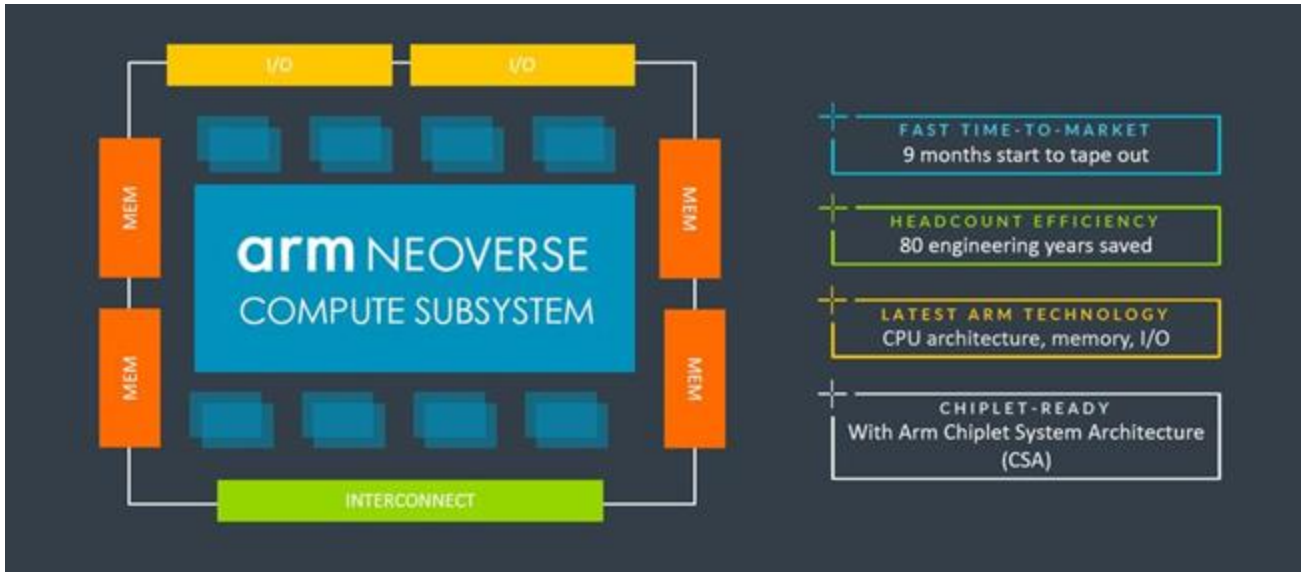



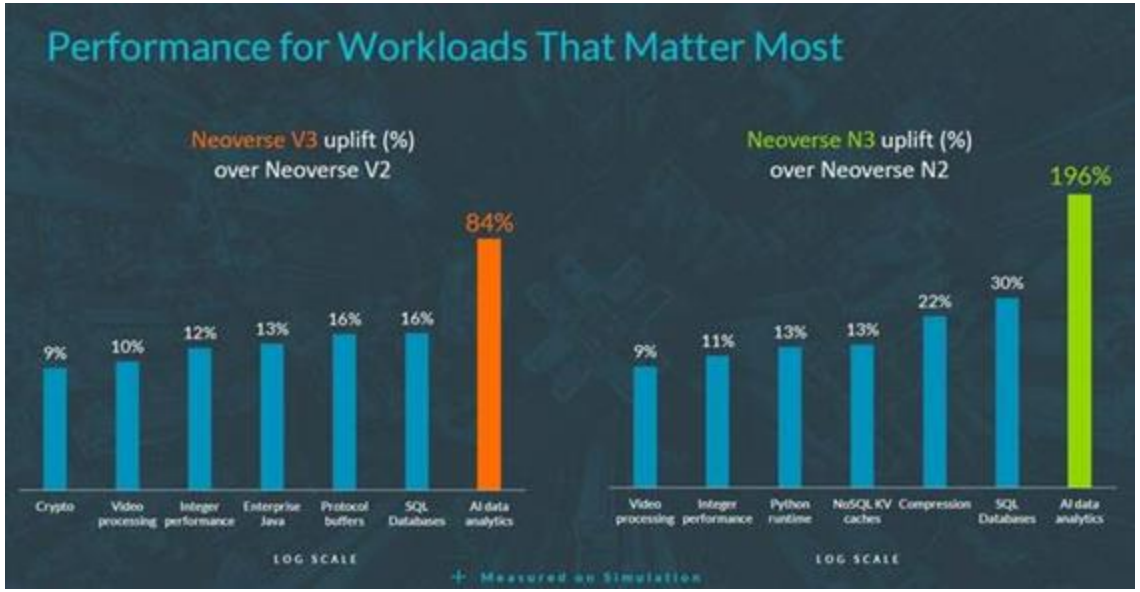



# Neoverse Reference Design (RD)

- Neoverse reference designs (RD) provide resources with best practices on how to integrate a Neoverse compute subsystem within a larger SoC. These compute subsystems are targeted at addressing requirements for applications in the cloud-to-edge infrastructure markets.
- Neoverse products are categorized as follows:
  - Neoverse V-Series: Maximum Performance
  - Neoverse N-Series: Scale Out Performance
  - Neoverse E-Series: Efficient Throughput
- The Neoverse reference designs are also available as Fixed Virtual Platform (FVP) models and when used with a software stack
  - Provide a way to explore the features available in the reference design compute subsystem through software.
  - Provide companion software stack for each of the Neoverse reference designs and the source code can be found in Arm's Gitlab repository (https://gitlab.arm.com/infra-solutions/reference-design)

# Arm Neoverse Compute Subsystem (CSS)





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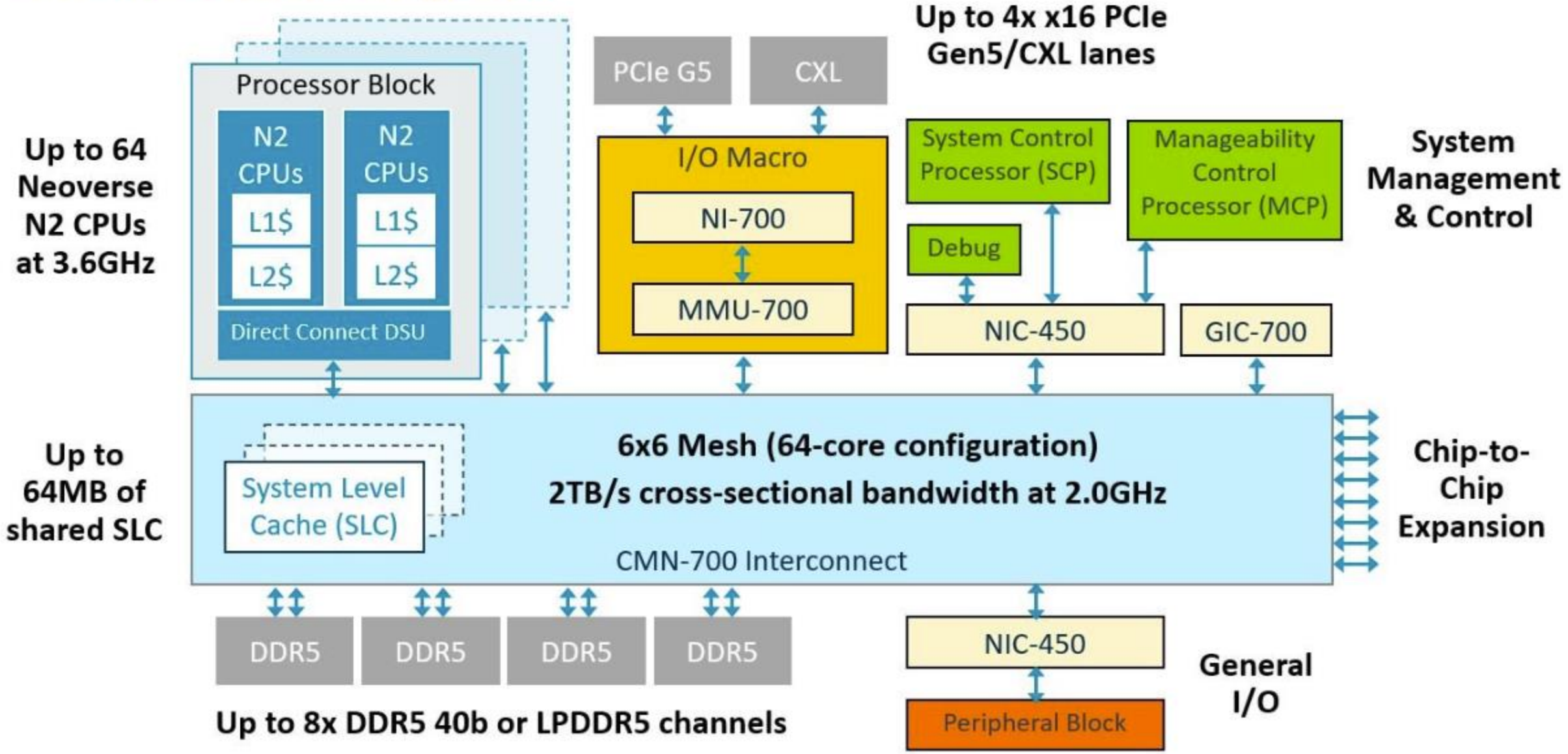
# CSS N2 Block Diagram

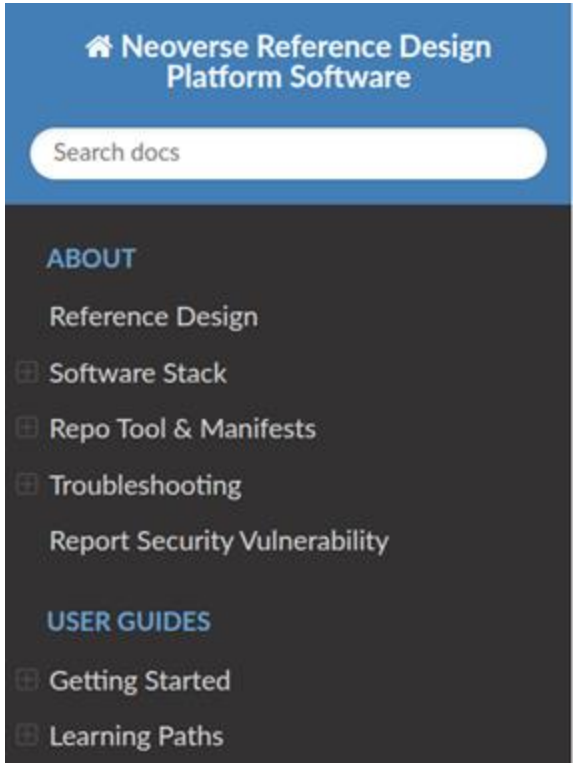


# Neoverse System IP

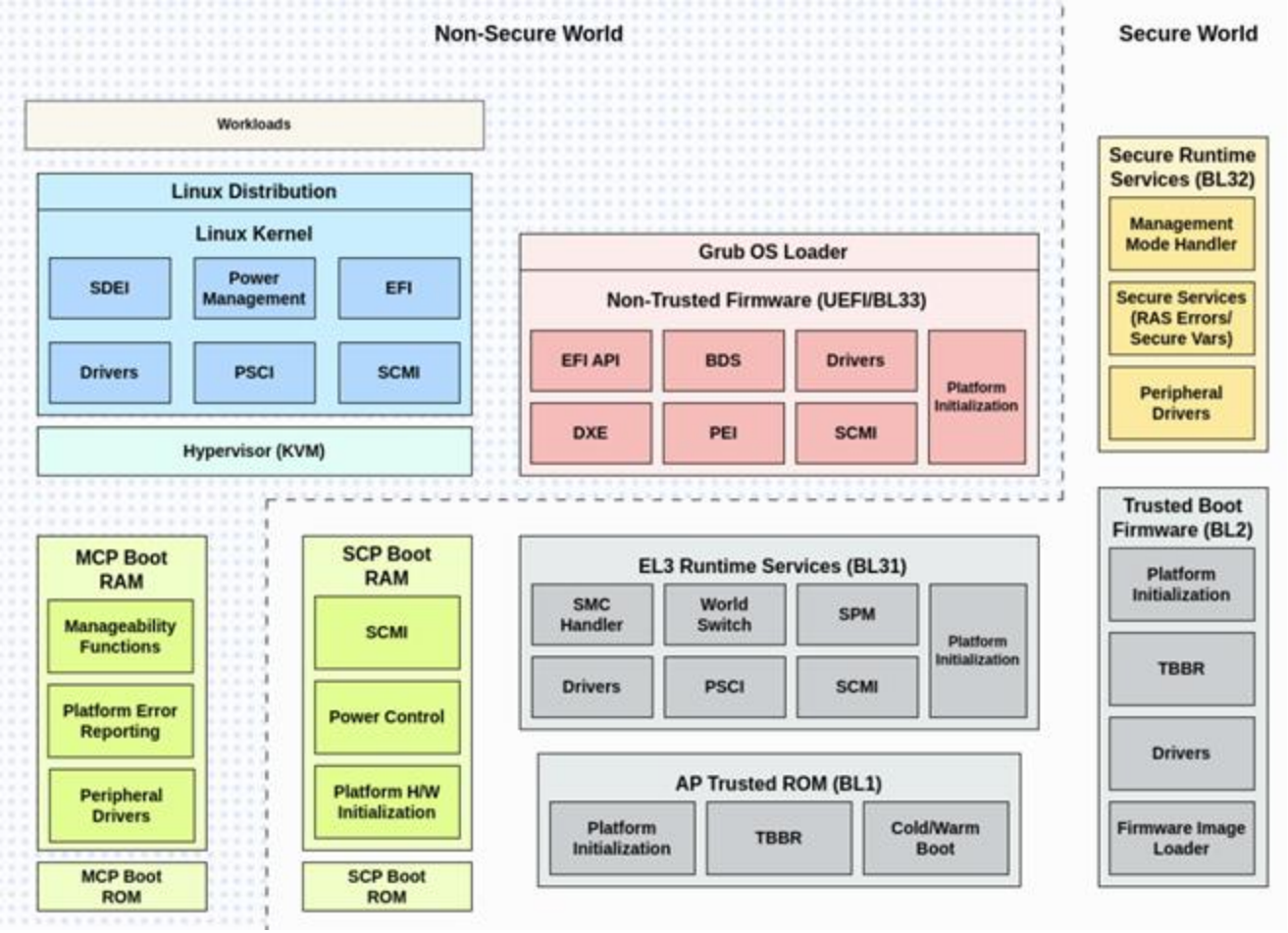
- Coherent Mesh Network Interconnect (CMN-700)
  - This is the main system interconnect infrastructure, connecting all the system components in a mesh topology. It contains the shared system level cache, and provides interfaces to on-chip and off-chip I/O, PCIe, on-chip DRAM, and die-to-die connectivity.
- Arm Generic Interrupt Controller (GIC-700)
  - supports the management and distribution of interrupts. It handles Shared Peripheral Interrupts (SPIs), Locality-Specific Peripheral Interrupts (LPIs), Private Peripheral Interrupts (PPIs), Message-Signalled Interrupts (MSIs), and Software-Generated Interrupts (SGIs).
- System Memory Management Unit (MMU-700)
  - In a virtualized system, each core in the system may have its own virtual address space, and each application running on each core may, in turn, have its own virtual address space.
  - Allows fully virtualized access to peripherals. The unit contains distributed address translation logic, which allows peripherals to be accessed using virtual addresses simultaneously in different contexts.
- Network Interconnect (NIC-700)
  - highly-configurable, high frequency, low latency on-chip system-level interconnect, capable of connecting across power and clock domain crossings.

# Neoverse Reference Design (RD) - Platform Software Stack

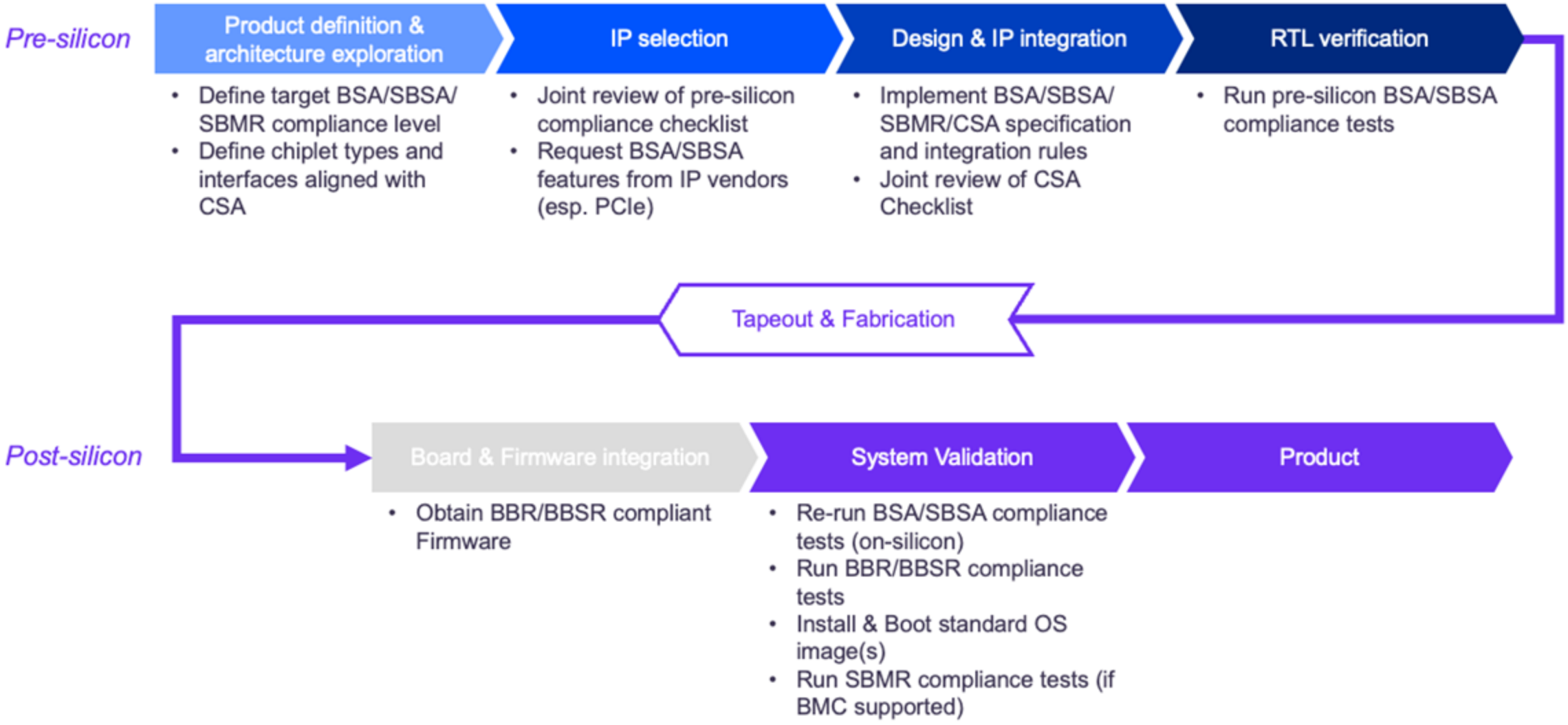
- The Neoverse software stack integrates multiple software components to provide a reference implementation of a software solution that can be used demonstrate various capabilities of the respective platform. A typical software stack is illustrated in below



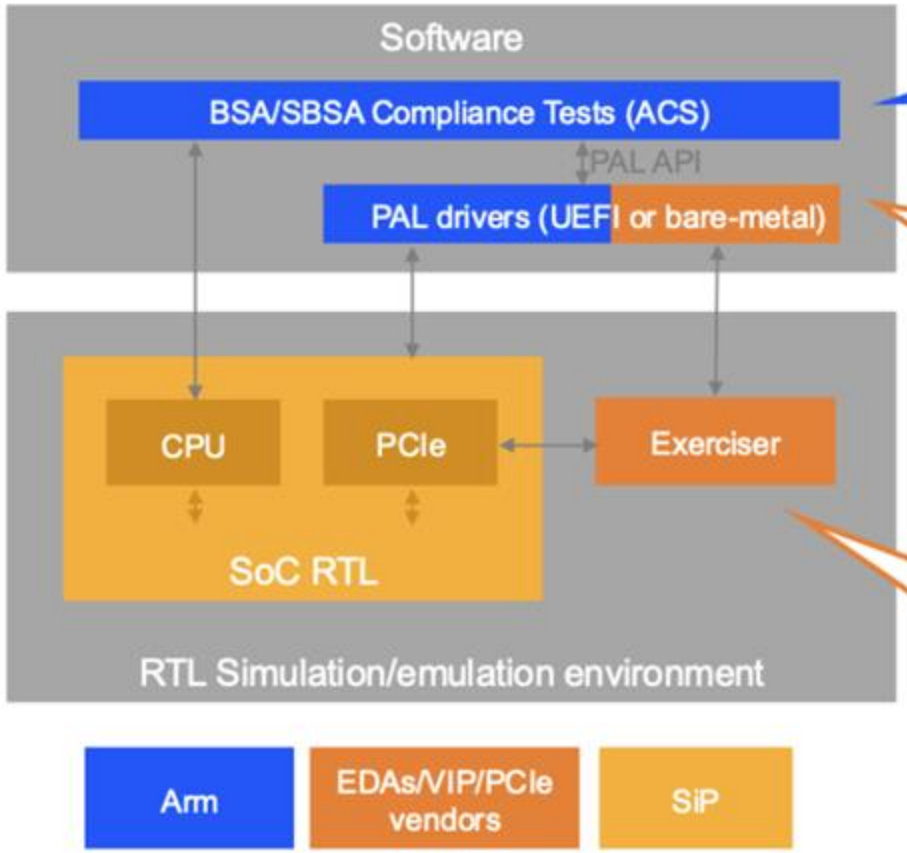

https://neoverse-reference-design.docs.arm.com/en/latest/about/software_stack.html



# Journey to product compliance



# Pre-Silicon Compliant Testing Suites




**Hardware Compliance tests**

- BSA-ACS for BSA compliance testing
  - https://github.com/ARM-software/bsa-acs
- SBSA-ACS for SBSA compliance testing
  - https://github.com/ARM-software/sbsa-acs

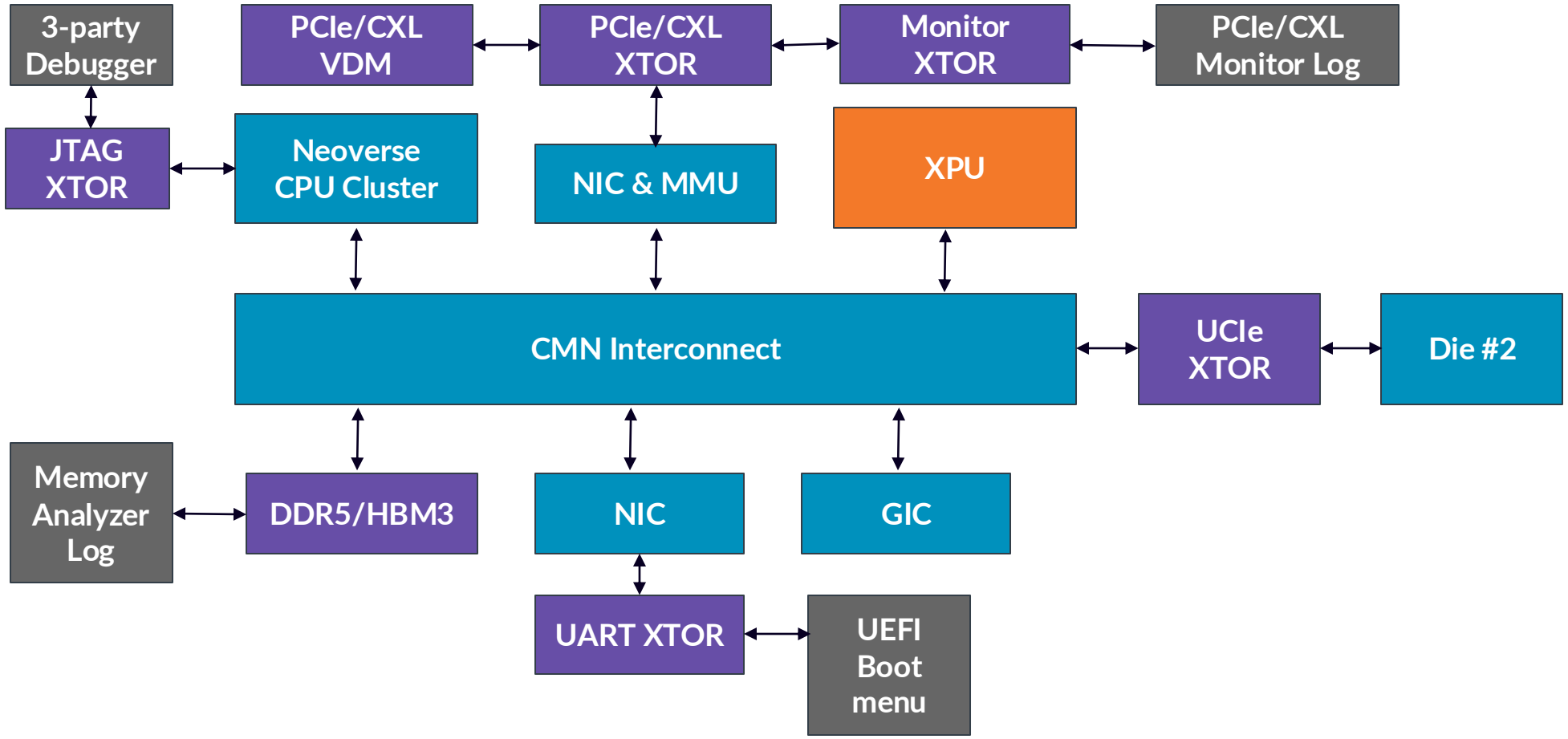
**UEFI or bare-metal driver options**

- **UEFI** - EL2 firmware (BL33/UEFI) with ACPI support
- **Bare-metal** – EL3 firmware
  - User guide - https://github.com/ARM-software/bsa-acs/blob/main/docs/arm_bsa_architecture_compliance_bare-metal_user_guide.pdf
  - Reference code (RD-N2 FVP) - https://github.com/ARM-software/bsa-acs/tree/main/pal/baremetal

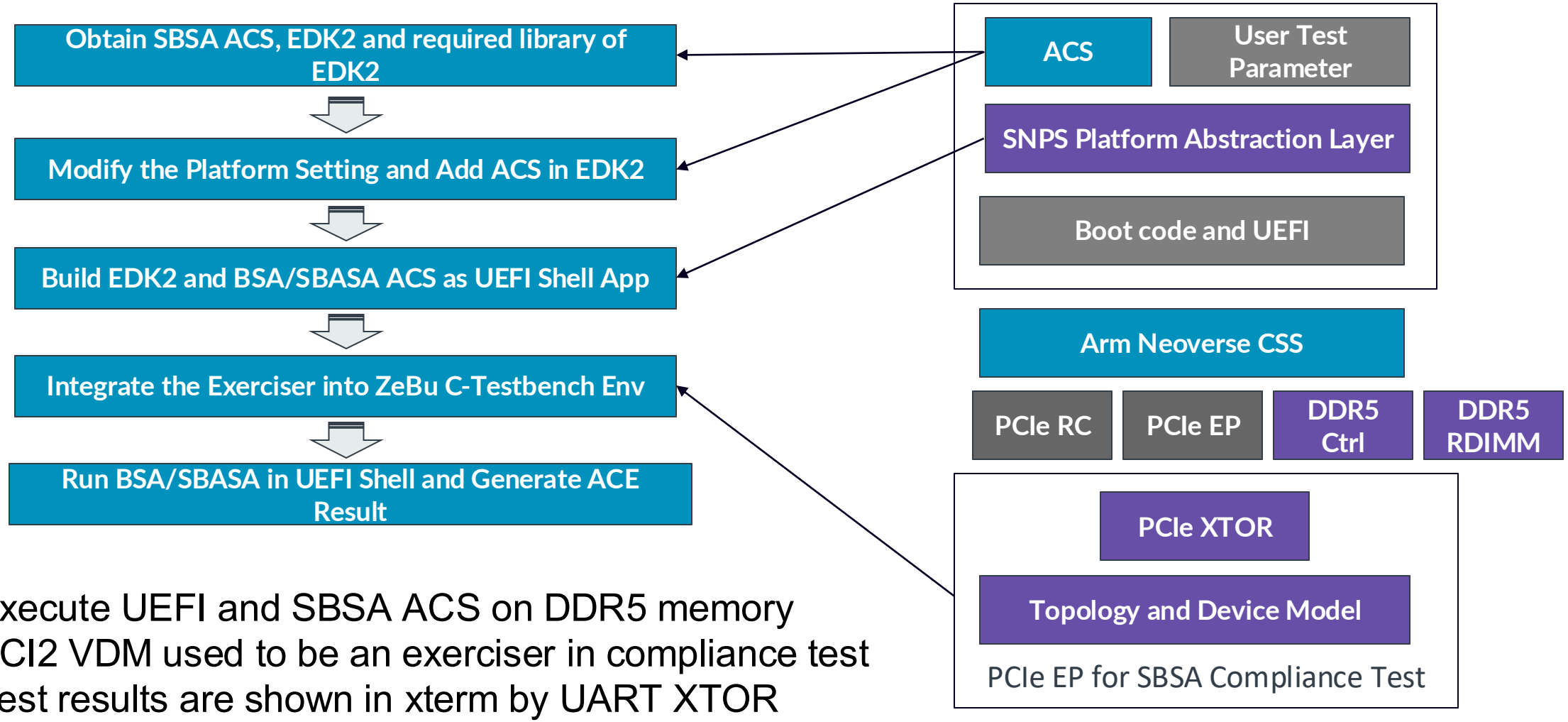
**Exerciser**

- Validation IP to cover the PCIe tests that requires End-point devices.

# Illustrated System Verification Architecture



# Running SBSA ACS UEFI-based Test in Emulation platform




- Execute UEFI and SBSA ACS on DDR5 memory
- PCI2 VDM used to be an exerciser in compliance test
- Test results are shown in xterm by UART XTOR

# Running ACS Testing on Neoverse-RDN2 FVP

FVP terminal_ns_uart_ap

```
Boot Manager

Boot Manager Menu

UEFI Shell
UEFI Non-Block Boot Device
UEFI Misc Device
UEFI Non-Block Boot Device 2
UEFI Non-Block Boot Device 3
UEFI Non-Block Boot Device 4

Device Path :
Fv(89CC2AB6-B847-475F-93E2-819603C3D15A)/FvFile(7C04A583-9E3E-4F1C-AD65-E05268D0B4D1)

Use the <^> and <v> keys to choose a boot option,
the <Enter> key to select a boot option, and the
<Esc> key to exit the Boot Manager Menu.

^v=Move Highlight      <Enter>=Select Entry      Esc=Exit
```

FVP terminal_ns_uart_ap

```
UEFI Interactive Shell v2.2
EDK II
UEFI v2.70 (EDK II, 0x00010000)
Mapping table
      FS0: Alias(s):F0:
          Fv(89CC2AB6-B847-475F-93E2-819603C3D15A)
      FS1: Alias(s):F1:
          MemoryMapped(0xB,0xE0000000,0xE01FFFFF)
      FS2: Alias(s):F2::BLK0:
          VenHw(5A96CDCD-6116-4929-B701-3AC2FB1CE228)
      FS3: Alias(s):F3::BLK1:
          VenHw(DE6AE758-D662-4E17-A97C-4C5964DA4C41,00)
     BLK2: Alias(s):
          VenHw(DE6AE758-D662-4E17-A97C-4C5964DA4C41,01)
Press ESC in 5 seconds to skip startup.nsh or any other key to continue.
Shell> FS2:
FS2:\> bsa.efi -f acs.txt
```

FVP terminal_ns_uart_ap

```
     *** Starting Exerciser tests ***

Operating System View:
 901 : Check P2P ACS Functionality
       Skipped on PE -    0
       PCI_PP_04
       Checkpoint --  1                          : Result:  SKIPPED
 902 : Check ACS Redirect Req Valid
       Skipped on PE -    0
       PCI_PP_04
       Checkpoint --  1                          : Result:  SKIPPED
 903 : Arrival order Check                       : Result:  PASS
 904 : MSI(-X) triggers intr with unique ID      : Result:  PASS
 905 : Generate PASID transactions          ConvertPages: Incompatible memory types, the pages to free have been freed
ConvertPages: Incompatible memory types, the pages to free have been freed
     : Result:  PASS
 906 : Generate PCIe legacy interrupt
       Skipped on PE -    0
       PCI_LI_02
       Checkpoint --  1                          : Result:  SKIPPED
 907 : Check PCIe I/O Coherency                  : Result:  PASS
 908 : Check PCIe Software Coherency             : Result:  PASS
 910 : Check RP Sec Bus transaction are TYPE0    : Result:  PASS
 911 : MSI to Any ITS Blk in assigned group      : Result:  PASS
 912 : MSI to ITS Blk outside assigned group     : Result:  PASS
 913 : MSI originating from different master     : Result:  PASS
 914 : P2P transactions must not deadlock        : Result:  PASS
 915 : Check ARI forwarding enable rule          : Result:  PASS
 916 : PCIe Memory access check                  : Result:  PASS
 917 : Check BME functionality of RP             : Result:  PASS

     One or more Exerciser tests failed or were skipped.

    ----------------------------------------------------
    Total Tests run  =  100  Tests Passed  =   81  Tests Failed =    2
    ----------------------------------------------------

     *** BSA tests complete. Reset the system. ***

remove-symbol-file /home/demo/labs/infra/fvp-rdn2/uefi_build/edk2/Build/Shell/DEBUG_GCC49/AARCH64/ShellPkg/Application/bsa-acs/uefi_app/BsaAcs/DEBUG/Bsa.dll 0xF96EC000
FS2:\>
```

Running ACS on Neoverse RDN2 FVP - demo video [Video]

# Simulate OpenBMC and UEFI pre-silicon on Neoverse RD-V3

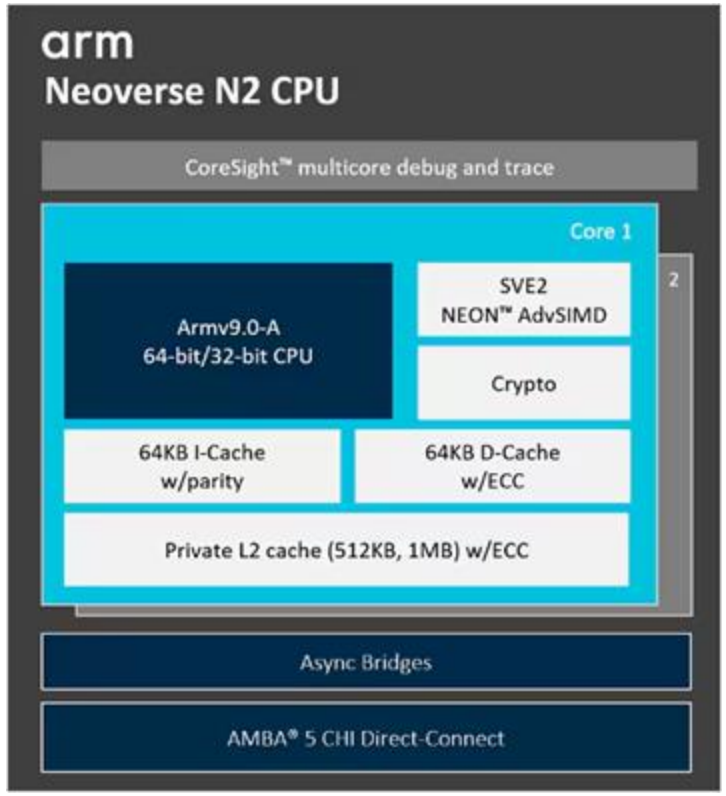
- OpenBMC
  - A collaborative open-source firmware stack for Baseboard Management Controllers (BMC), hosted by the Linux Foundation. BMCs are embedded controllers on server motherboards that enable both in-band and out-of-band system management. Out-of-band access allows remote management even when the host is powered off or unresponsive, while in-band interfaces support communication with the host operating system during normal operation.
- Key interactions between OpenBMC and UEFI
  - The Unified Extensible Firmware Interface (UEFI) is the modern replacement for legacy BIOS, responsible for initializing hardware and loading the operating system. UEFI provides a robust, modular, and extensible interface between platform firmware and OS loaders.

| Interaction | Direction | Description |
|---|---|---|
| Boot power sequencing | BMC → Host | BMC controls host power-on flow, ensuring UEFI starts in the correct sequence |
| Boot status reporting | UEFI → BMC | UEFI reports boot state and progress via IPMI (KCS) or PLDM |
| Serial over LAN (SOL) | BMC ↔ Host | BMC bridges the host UART console to remote clients over the network |
| Pre-boot configuration | BMC ↔ UEFI | BMC may inject or read boot settings via shared memory or commands |
| System recovery signaling | UEFI → BMC | UEFI can request BMC-initiated reboot, NMI, or recovery actions |

https://learn.arm.com/learning-paths/servers-and-cloud-computing/openbmc-rdv3/

# Build a RAG Pipeline on Arm Neoverse Server

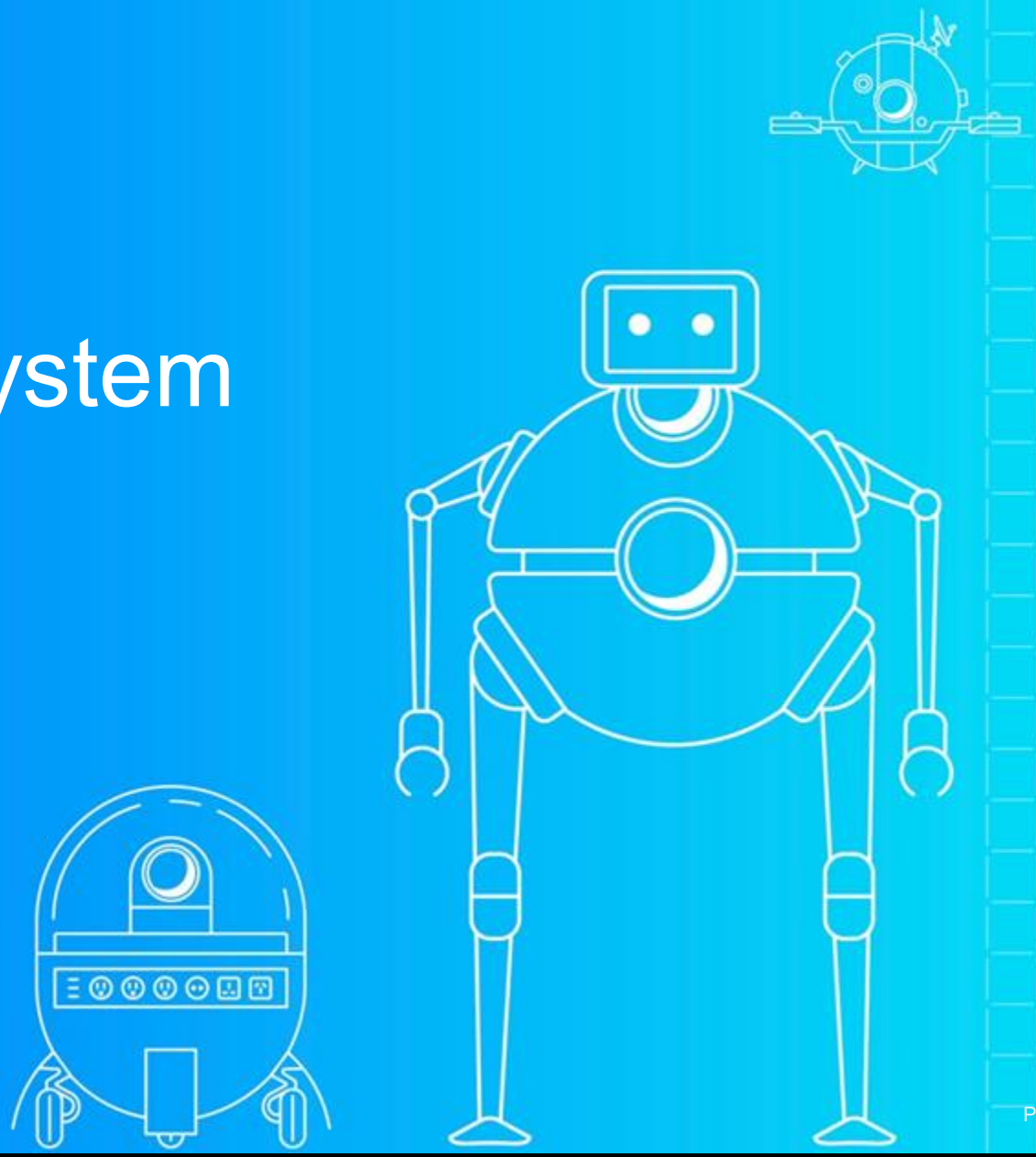
- Arm Neoverse evaluation kits
  - 24-core Neoverse-N2 edge server, 48GB ECC DDR5,
  - Ubuntu 22.04 OS
  - Gemma3-27B LLM Chatbot at 10 tokens/s
- Build a RAG pipeline on Arm-based server
  - Instead of relying solely on pre-trained weights, a Retrieval-Augmented Generation (RAG) system retrieves relevant text from a document corpus and passes it to a language model to create factual, context-aware responses.
- Transforming a question into a context-aware response
  - Embedding model: Converts text into dense numerical vectors. An example is e5-base-v2.
  - Vector database: Searches for semantically similar chunks. An example is FAISS (Facebook AI Similarity Search).
  - Language model: Generates an answer conditioned on retrieved context. An example is Llama 3.1 8B Instruct.
  - Start the llama.cpp REST server before running the RAG query.

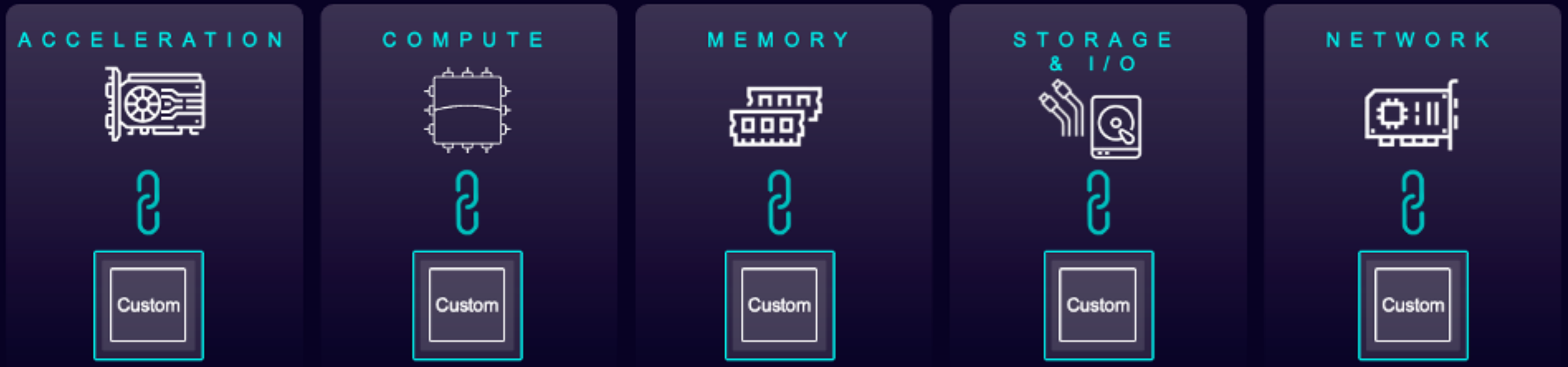


https://learn.arm.com/learning-paths/laptops-and-desktops/dgx_spark_rag/

arm

# Enabling Chiplet Ecosystem with Arm Total Design






# CUSTOM SILICON

Constraint-specific optimization

Co-designed for max performance per watt

Deep feature integration

# Arm is Driving Cross-industry Collaboration to Accelerate Custom Silicon Design for AI

Pre-integrated IP
Validated EDA Tool Flows

Advanced Foundry &
Packaging Support

Expert Design
Services

Commercial Software
& Firmware for Servers

AI Systems
(Racks)

Accelerator
(GPU)

Compute
(CPU)

## NVIDIA

VERA RUBIN AI CLUSTER

Rubin

Vera

## aws

AI ULTRACLUSTER

Trainium

Graviton4

## Google Cloud

CLOUD TPU RACK

TPU

Axion

## Microsoft

AZURE AI RACK

Maia

Cobalt

# Arm Neoverse: Optimal Performance & Efficiency

| aws | Google Cloud | Microsoft Azure | Alibaba Cloud |
|---|---|---|---|
| Graviton | Axion | Cobalt100 | Yitian 710 |
| 60% | 60% | 40% | 50% |
| more power efficient over comparable EC2 instances | more power efficient over comparable x86 instances | more power efficient over previous generation | more power efficient over comparable Arm servers |

Seamless Developer and User Experience across Multicloud

# Arm Total Design Ecosystem - Accelerate Your Custom Silicon

- Arm Total Design provides partners with preferential access to Arm Neoverse CSS, pre-integrated IP and EDA tools, design services, foundry support, and commercial software and firmware support. By working with this ecosystem, partners can leverage the expertise and resources of the industry to bring custom silicon solutions rapidly to market.

### Robust Third-Party IP

Pre-integrated, validated IP and EDA tools from partners such as Cadence, Rambus, and Synopsys, help accelerate silicon design and feature integration, including memory, security, and peripherals.

### Expert Design Services

Design services from partners including ADTechnology, Alphawave Semi, Broadcom, Capgemini, Faraday, Socionext, and Sondrel, who are ready to support the ecosystem with expertise on Neoverse CSS, other Arm Total Design IP, and methodology.

arm TOTAL DESIGN

### Advanced Foundry Support

Technology optimized for leading-edge process nodes and advanced packaging techniques from foundry partners, including Intel Foundry Services and TSMC.

### Commercial Software & Firmware

Commercial software and firmware support for Neoverse CSS from leading infrastructure firmware providers, including AMI.

# arm total design

## FOUNDRY SERVICES

- intel foundry services
- SAMSUNG
- tsmc

## DESIGN SERVICES & PACKAGING

- ADTechnology
- alchip
- ASE
- ALCOR MICRO
- asicland
- BROADCOM
- CoAsia
- GUC
- FARADAY
- HCLTech
- MARVELL
- MEDIATEK
- NOVATEK
- REALTEK
- SEMI FIVE
- socionext
- VIA NEXT

## FIRMWARE, EDA & 3RD PARTY IP

- ami
- cadence
- CEVA
- insyde
- proteanTecs
- Rambus
- SYNOPSYS
- PUFsecurity
- SIEMENS

## COMPANION SILICON & OEMs

- AsteraLabs
- CREDO we connect.
- eliyan
- Jaguar
- NOKIA
- rebellions_
- SIPEARL

New Members in 2025

## Linux/OS
(Open Source & Commercial)

Red Hat Enterprise Linux, fedora, Canonical, openSUSE, debian, alpine, AlmaLinux, FreeBSD, CentOS, Rocky Linux, ORACLE Linux, ...

## Cloud Native
(Open Source & Commercial)

kubernetes, Istio, CNI, docker, fluentd, GitHub, dapr, LINKERD, envoy, falco, NGINX+, JFrog, elastic, etcd, ...

## SaaS & Enterprise

Adobe, airbnb, SAP, Spotify, Uber, CROWDSTRIKE, splunk>, mongoDB., FORTINET, EPIC, METRO WEATHER, rescale, templafy, SIEMENS, Paramount+, 8x8, dynatrace, DATADOG, paloalto, ORACLE, CLOUDFLARE, snowflake, ...

## AI/ML

PyTorch, ONNX, ANACONDA, LLAMA.CPP, Hugging Face, zilliz, ANTHROP\C, CLEARML, databricks, arcee.ai, ...

# Arm Neoverse Compute Platform

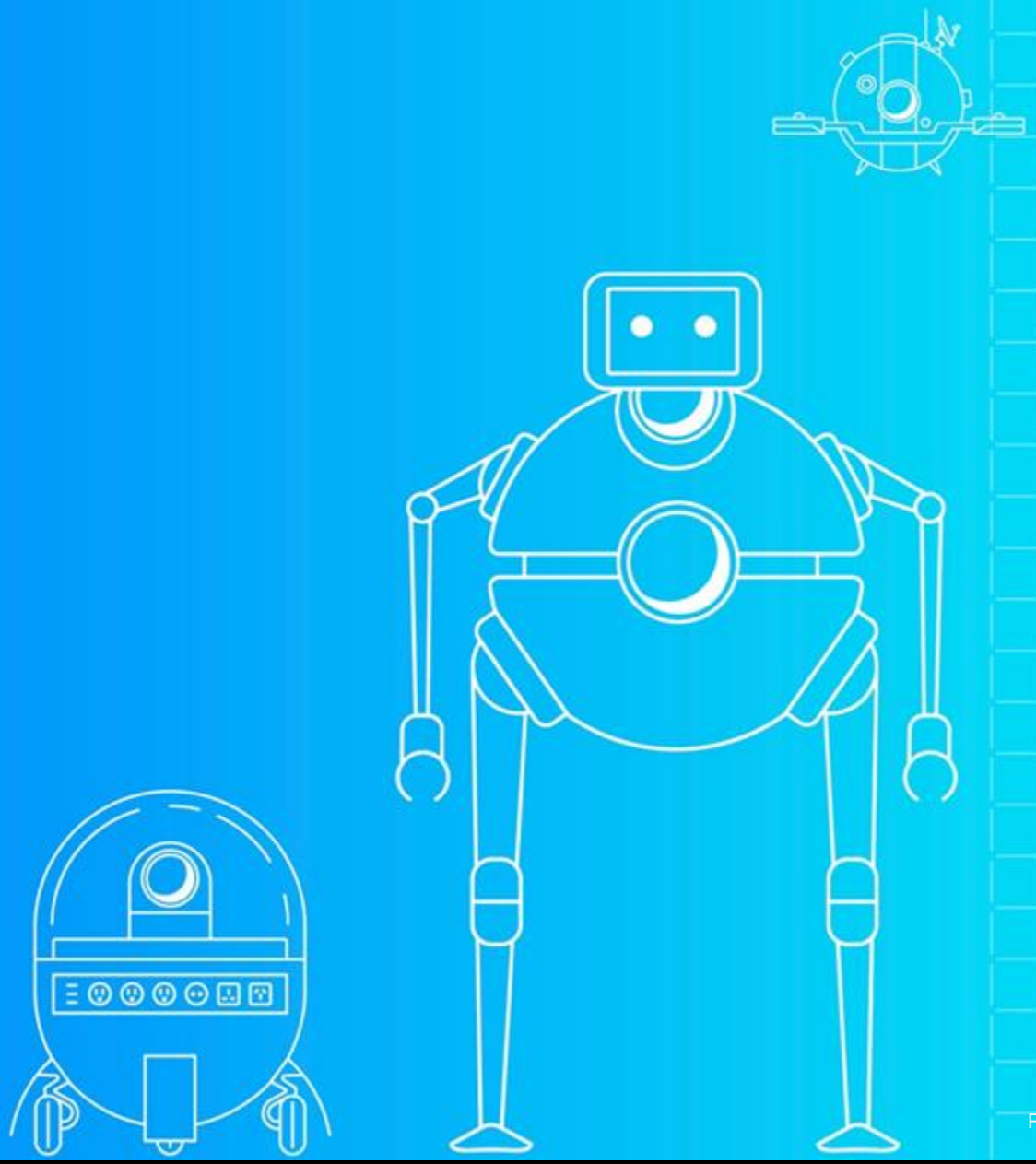
## Leveraged Software Investment across Arm Silicon

# Software Ecosystem Dashboard for Arm

For AI, Cloud, Data Center, 5G, Networking and Edge Software Packages

https://developer.arm.com/ecosystem-dashboard/

# Summary



# Summary and Takeaway

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- Learning Resource
  - Introduction-to-Neoverse [Arm Education Github]
  - Neoverse RD-N2 Platform: RD-INFRA-2024.12.20 [Arm Gitlab]
  - Neoverse Reference Design Platform Software [Arm Document]
  - What are OpenBMC and UEFI? [Arm Learning Path]
  - Simulate OpenBMC and UEFI pre-silicon on Neoverse RD-V3 [Arm Learning Path]
  - Optimizing Generative AI on Arm Processors [Arm Education Github]
  - Build a RAG pipeline on Arm-based NVIDIA DGX Spark [Arm Learning Path]

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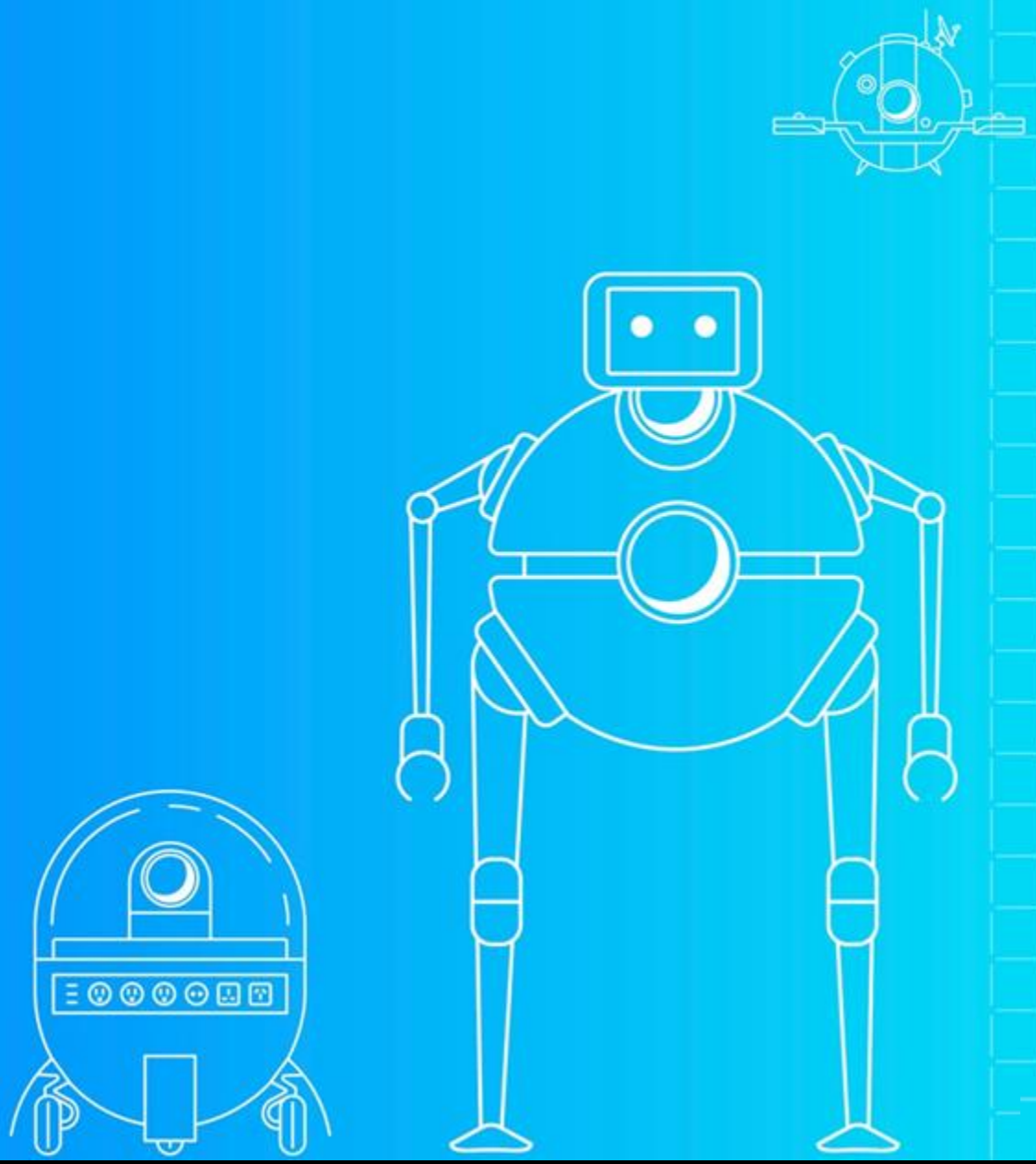
# Q & A